Вкладення Йонеди
в сімпліціальній теорії типів

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Анотація

Ріль і Шульман [46] ввели *симпліціальну теорію типів* (STT), варіант гомотопічної теорії типів, який мав на меті вивчати не лише теорію гомотопій, але і її злиття з теорією категорій: теорію $(\infty, 1)$-категорій. Хоча це і є технічно складним, маніпулювання ∞-категоріями в симпліціальній теорії типів часто є простішим, ніж робота зі звичайними категоріями, оскільки теорія типів обробляє нескінченні стеки когерентностей у фоновому режимі. Ми спираємося на нещодавню роботу [19], що визначає $(\infty, 1)$-категорію ∞-групоїдів у STT, щоб визначити категорії передпучків у межах STT та систематично розвинути їхню теорію. Зокрема, ми будуємо вкладення Йонеди, доводимо універсальну властивість категорій передпучків, уточнюємо теорію спряженостей у STT, вводимо теорію розширень Кана та доводимо теорему A Квіллена. На додаток до великого обсягу теорії категорій у STT, ми пропонуємо вагомі докази того, що STT може бути використана для отримання складних результатів у теорії ∞-категорій з набагато меншою складністю.

Присвячується світлій пам'яті Томаса Штрайхера (1958–2025)

1 Вступ

Расселл [48] відомим чином описав два стилі формалізації математики як різницю між *крадіжкою* та *чесною працею*. Обидва підходи можна побачити в сучасному використанні залежної теорії типів. Чесна праця передбачає *аналітичний* підхід: трактування типів як базових об'єктів, еквівалентних множинам, а також визначення і міркування про такі об'єкти, як дійсні числа, групи та топологічні простори, як це робиться зазвичай. Це те, що робиться, наприклад, у доведенні Coq Теореми про непарний порядок [14]. Більш швидкий шлях крадіжки передбачає розгляд теорії типів як спеціалізованої *синтетичної* мови для певного типу математичних об'єктів і постулювання їх базових властивостей. Це звужує сферу застосування теорії типів, але, водночас, робить доведення про ці конкретні об'єкти набагато лаконічнішими. Наприклад, *гомотопічна теорія типів* (HoTT) [53] постулює різні аксіоми, які гарантують, що типи поводяться як простори (з точністю до гомотопії), уможливлюючи доведення теорем з алгебраїчної топології без введення явного опису простору. Насправді, синтетичний підхід менше схожий на крадіжку, ніж на позику; ви розплачуєтеся за налаштовану теорію типів семантичною моделлю, яка інтерпретує типи як потрібні об'єкти і валідує додаткові аксіоми.

У цій роботі ми використовуємо синтетичну методологію для застосування теорії типів до вивчення теорії категорій. Зокрема, ми додаємо різні аксіоми до гомотопічної теорії типів, щоб побудувати систему, де гасло HoTT "всі типи є просторами, а всі функції неперервними" заміняється на "(деякі) типи є (∞-)категоріями, а всі функції є функторами".[1] Це розширення теорії типів називається *симпліціальною* теорією типів (STT) і було введено [46].

Хоча знання ∞-категорій не є обов'язковим для використання нашої теорії, груба інтуїція щодо них допомагає для розуміння STT. Тому ми нагадаємо наступне нечітке визначення. ∞-категорія C – це сукупність об'єктів з *простором* стрілок між об'єктами c та d, $\mathsf{hom}(c, d)$, а не множиною, оснащена неперервною операцією композиції та призначенням тотожних стрілок. Вирішальним є те, що операція композиції має бути асоціативною та унітальною лише з точністю до гомотопії, але з обмеженням, що ці гомотопії самі задовольняють закони *когерентності* у вигляді додаткових гомотопій, і так далі з когерентностями між когерентностя-

[1] У цій статті під ∞-категорією ми розуміємо $(\infty, 1)$-категорію.

ми тощо. Як вільна аналогія, подібно до того, як моноїдальна категорія послаблює моноїди, дозволяючи $\otimes$ бути асоціативним з точністю до ізоморфізмів, що задовольняють певні рівняння когерентності, ∞-категорії послаблюють звичайні категорії дозволяючи законам категорій виконуватися лише з точністю до (нескінченно когерентних) ізоморфізмів.

Прикметно, що практично кожна теорема, на яку можна сподіватися для звичайних категорій, виконується для ∞-категорій.[2] Однак, доведення є набагато складнішими, оскільки вони працюють з *моделями* ∞-категорій (інструментами, що використовуються для організації та управління вежею когерентностей [5]). Метою STT є використання теорії типів, щоб приховати когерентності від користувача і дозволити доведення, які є не складнішими, ніж класичні міркування для 1-категорій.

У цій роботі ми надаємо вагомі докази цієї гіпотези, розвиваючи велику частину теорії категорій—кілька основних результатів *Категорій для працюючого математика* [28]—виключно в межах STT.

1.1 Симпліціальна теорія типів

Щоб побудувати теорію типів для синтетичної теорії категорій, можна було б сподіватися інтерпретувати теорію типів у категорію категорій (∞ або інакше), щоб гарантувати, що типи реалізують категорії. Однак, категорія CAT малих категорій поводиться занадто погано, щоб утворити модель теорії типів Мартіна-Льофа (MLTT). Замість цього, [46] розширюють CAT і вкладають її як рефлективну підкатегорію в категорію (∞-)передпучків на симплекс-категорії $\widehat{\Delta}$, яка є достатньо багатою для моделювання HoTT. Тоді STT аксіоматизує частину $\widehat{\Delta}$, щоб ізолювати CAT як рефлективний підуніверсум у межах теорії типів [47].

Ми введемо повний набір доповнень у Розділі 2 (зібраних у Додатку Б для зручності), але найважливішим серед них є постульований *інтервальний тип* $\mathbb{I} : \mathcal{U}_0$. Ми додатково припускаємо, що $\mathbb{I}$ є обмеженим лінійним порядком з кінцевими точками $0, 1 : \mathbb{I}$. Інтуїтивно, $\mathbb{I}$ має фіксувати категорію $\{0 \to 1\}$—він так інтерпретується в $\widehat{\Delta}$—і ми можемо використовувати це для визначення та дослідження типу *синтетичних морфізмів* у довільному типі X: стрілка в X відповідає звичайній функції $\mathbb{I} \to X$, причому обчислення в $0, 1$ дає домен і кодомен. Наприклад,

[2] Принаймні, як зауважив один з наших рецензентів, за умови правильного калібрування своїх сподівань.

тотожна стрілка в $x : X$ задається як $\lambda\_.\, x$.

Однак, подібно до того, як цільова модель $\widehat{\Delta}$ є строго більшою за CAT, не всі типи в STT достовірно моделюють категорії. Зокрема, хоча завжди можна побудувати тотожні морфізми, не всі типи мають оператор композиції. Дивовижно, проте, що оператори композиції є унікальними, коли вони існують, і їх існування для типу X фіксується відносно коротким твердженням (Визначення 17). Маючи під рукою операцію композиції для X, ми можемо визначити тип ізоморфізмів у X, і ми визначаємо категорію як тип, де (1) операція композиції існує унікально з точністю до гомотопії, і (2) тип ізоморфізмів у X еквівалентний типу ідентичності $=_X$.

***Remark* 1.** Цей останній пункт принципово залежить від *відмови* від припущення унікальності доведень ідентичності, щоб ми випадково не заборонили будь-якій синтетичній категорії мати об'єкт з нетривіальним автоморфізмом. Однак, припускаючи, що ізоморфізми та доведення ідентичності збігаються, ми можемо використовувати підтримку теорією типів заміни рівного на рівне для безперешкодного перенесення доведень уздовж ізоморфізмів. Ось чому робота з HoTT/інтенсіональною теорією типів при формулюванні синтетичної теорії категорій виявляється зручнішою, ніж екстенсіональна теорія типів, навіть якщо хтось не цікавиться ∞-категоріями.

1.2 Теорія категорій всередині STT

Хоча деякі нещодавні роботи досліджували STT для її застосувань у мовах програмування [60, 19, 56], більшість робіт із симпліціальної теорії типів були зосереджені на доведенні результатів з теорії категорій всередині теорії типів [46, 44, 4, 57, 10, 59, 58]. З цією метою теорія спряженостей, дискретних розшарувань та розшарувань Гротендіка, а також (ко)границь були введені та вивчені в межах симпліціальної теорії типів. Деякі з цих результатів, наприклад, лема Йонеди для розшарувань [46], були згодом механізовані [23].

Донедавна, однак, у STT не існувало замкнених типів, які б представляли нетривіальні категорії. Як наслідок, хоча [46] і наводить відмінне визначення спряженостей, жодних прикладів навести не можна. У попередній роботі ми змінили це, розширивши STT для побудови $\mathcal{S}$, категорії $\mathcal{S}$просторів, яка є гомотопічним аналогом SET [19]. Об'єкти $\mathcal{S}$

– це елементи $\mathcal{U}_0$, які кодують ∞-групоїди, а морфізми в $\mathcal{S}$ відповідають їх функціям. *Там само* $\mathcal{S}$ використовується як будівельний блок для відновлення алгебраїчних категорій (груп, кілець), а також інших прикладів (частково впорядкованих множин, симплекс-категорії тощо).

Наше розширення STT використовувало різні *модальності* поверх HoTT для побудови $\mathcal{S}$. Тут ми беремо $\mathcal{S}$ повністю, але деякі з використаних нами модальностей все ще є критично важливими для формулювання природних теорем у теорії категорій. Відповідно, у цій статті ми також працюємо в межах модального розширення HoTT, заснованого на MTT [15].

1.3 Внесок

Ми переглядаємо базову теорію категорій у світлі побудови $\mathcal{S}$ і показуємо, що більшість класичних результатів, з якими стикаються в теорії категорій, тепер знаходяться в межах досяжності симпліціальної теорії типів. Вперше ми показуємо, що STT може бути використана для доведення життєво важливих теорем у теорії ∞-категорій без звернення до складних моделей. Багато з цих теорем (наприклад, цілком лояльні та суттєво сюр'єктивні функтори є еквівалентностями) прямо не згадують $\mathcal{S}$, але критично спираються на принципи міркування, які уможливлює $\mathcal{S}$. Ми доводимо два робочі результати для категорій передпучків $\widehat{C}$:

- Ми будуємо цілком лояльну функцію $\mathsf{Yo} : C \to \widehat{C}$.
- Ми доводимо, що $\widehat{C}$ є "вільним копоповненням C".

Ключовою технічною інновацією для цього є категорія *скручених стрілок* (twisted arrow), яку ми інтегруємо в STT як модальність. Після цього ми можемо вивести різні класичні результати, наприклад:

- що поточково оборотні відображення в $C \to D$ є оборотними;
- що поточково ліві спряжені є лівими спряженими;
- що (ко)границі обчислюються поточково в $C \to D$;
- теорію та існування поточкових розширень Кана;
- теорему A Квіллена;

- власніть кодекартових розшарувань.

Синтетичний підхід дає лаконічне формулювання:

Axiom A. *Існує множина* $\mathbb{I}$*, яка утворює обмежену дистрибутивну решітку* $(0, 1, \vee, \wedge)$*, таку що виконується* $\prod_{i,j:\mathbb{I}} i \leq j \vee j \leq i$*.*

Ми розглядаємо $\mathbb{I}$ як *спрямований інтервал*, і [46] використовують його, щоб оснастити кожен тип поняттям синтетичного морфізму:

Definition 2. Синтетичний морфізм $f : \mathsf{hom}_X(x, y)$, де $x, y : X$, — це функція $f : \mathbb{I} \to X$ разом із пропозиційними рівностями $f\,0 =_X x$ та $f\,1 =_X y$.

***Remark* 3.** У [46] синтетичний інтервал визначається як більш примітивна дедуктивна структура, і $\mathsf{hom}(x, y)$ використовує типи строгого розширення. Це дає більше дефініційних рівностей: $f\,0$ та x збігалися б дефініційно, коли $f : \mathsf{hom}(x, y)$ (і аналогічно для $f\,1$ та y). Однак дедуктивний підхід не включає безпосередньо $\mathbb{I}$ як звичайний тип, а його взаємодія з модальностями (Розділ 2.4) є складною. З цих причин ми працюємо з простішим, але менш строгим визначенням $\mathsf{hom}(x, y)$. У системі, де доступні обидва підходи, ці два поняття є еквівалентними [10].

Будь-яка функція $f : X \to Y$ автоматично має дію на синтетичні морфізми $\alpha : \mathbb{I} \to X$ через пост-композицію $f \circ \alpha : \mathbb{I} \to Y$. У цьому випадку ми часто пишемо $f(\alpha)$.

З $\mathbb{I}$ ми негайно отримуємо n-куби $\mathbb{I}^n$, а з них ми можемо виділити симплекси Δ^n, границі $\partial\Delta^n$ та роги Λ^n_k. Зокрема, $\Delta^2 \to X$ представляє 2-клітину в X, що свідчить про композицію двох стрілок, а $\Lambda^2_1 \to X$ представляє пару компонованих стрілок (без композиції). Нижче ми наведемо визначення цих типів:

$$\Delta^n = \{(i_1, \ldots, i_n) : \mathbb{I}^n \mid i_1 \geq i_2 \geq \cdots \geq i_n\}$$

$$\Lambda^2_1 = \{(i, j) : \mathbb{I}^2 \mid i = 1 \vee j = 0\}$$

Notation 4. Ми пишемо $i : \Delta^n$ $(0 \leq i \leq n)$ як скорочення для послідовності з i копій 1, за якими йдуть 0: $(1, 1, \ldots, 0, \ldots)$.

Відображення $f : \Delta^2 \to X$ називається свідченням того, що композиція $f(-, 0)$, за якою йде $f(1, -)$, є $\lambda i.\, f(i, i)$. Ми наголошуємо, що це — *дані*; може бути багато різних f, що свідчать про одну і ту саму композицію, оскільки X може мати багато нееквівалентних 2-клітин з однією

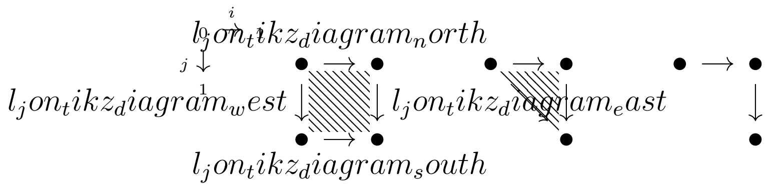

Рис. 1: Візуалізація $\mathbb{I}^2$, Δ^2 та Λ^2_1.

і тією ж границею. З цієї ж причини, однак, не завжди пара компонованих морфізмів $\Lambda^2_1 \to X$ поширюється до даних композиції $\Delta^2 \to X$. Це саме тому, що не кожен тип у STT можна розглядати як категорію; хоча ми визначили $\mathsf{hom}_X(x, y)$ для кожного X, не існує апріорного способу *компонування* цих морфізмів. Прекатегорії — це типи, для яких існують усі композиції:

Definition 5. *Прекатегорія* — це тип X, що задовольняє умову Сегала: вкладення $\Lambda^2_1 \to \Delta^2$ індукує еквівалентність $\mathsf{isEquiv}(X^{\Delta^2} \to X^{\Lambda^2_1})$.

Грубо кажучи, умова Сегала гарантує, що кожна пара компонованих морфізмів в X поширюється (унікально) до 2-клітини, що свідчить про їх композицію, і, зокрема, існує індукована функція композиції $\mathsf{hom}(x, y) \times \mathsf{hom}(y, z) \to \mathsf{hom}(x, z)$. Унікальність автоматично гарантує, що ця операція є асоціативною та унітальною. Визначення категорії уточнює це дещо далі. У прекатегорії X ми можемо визначити тип ізоморфізмів $x \cong y$ між $x, y : X$, і тому існує два потенційно різних типи свідчень того, що x та y є ідентичними: $x =_X y$ та $x \cong_X y$. Категорія — це прекатегорія, для якої ці два типи є канонічно еквівалентними.

Definition 6. $\alpha : \mathsf{hom}(x, y)$ є ізоморфізмом $(\mathsf{isIso}(\alpha))$, якщо існують $\beta_0, \beta_1 : \mathsf{hom}(y, x)$, такі що $\beta_0 \circ f = \mathsf{id}$, $f \circ \beta_1 = \mathsf{id}$.[3] Ми пишемо $\mathsf{Iso}(x, y)$ або $x \cong y$ для підтипу ізоморфізмів.

Definition 7. Прекатегорія C є *категорією*, якщо вона задовольняє умову Резка:

$$\prod_{x,y:C} \mathsf{isEquiv}(\mathsf{idtoiso} : (x = y) \to \mathsf{Iso}(x, y))$$

де $\mathsf{idtoiso}(\mathsf{refl}) := \mathsf{id}$. Якщо кожен морфізм у C є ізоморфізмом, то C є *групоїдом.*

Example 8. $\mathbb{I}, \Delta^n, \mathbb{I}^n$ є категоріями [19].

[3]Це саме HoTT-еквівалентність, але перенесена на синтетичні морфізми.

2 Модальна та симпліціальна теорія типів

У цій статті ми сприймаємо STT значною мірою як належне і зосереджуємося на роботі всередині теорії. Однак, щоб зробити цю статтю більш самодостатньою, ми присвячуємо цей розділ ретельному поясненню нових конструкцій модальної гомотопічної теорії типів та аксіом, що її доповнюють і утворюють симпліціальну теорію типів.

2.1 Гомотопічна теорія типів

Ми починаємо з нагадування основних понять і позначень гомотопічної теорії типів, які ми використовуємо в цій статті. Канонічним джерелом є книга HoTT [53]. Ми працюємо в межах інтенсіональної теорії типів Мартіна-Льофа і зауважимо, як HoTT розширює її.

Notation 9. Ми пишемо $a =_A b$ для типу ідентичності (часто опускаючи A). Дано $p : a =_A b$ та $B : A \to \mathcal{U}$, ми пишемо $p_!$ для відображення $B(a) \to B(b)$.

Definition 10. Ми кажемо, що функція $f : A \to B$ є еквівалентністю, якщо f допускає як лівий, так і правий обернений:

$$\mathsf{isEquiv}(f) = \textstyle\sum_{g,h:B\to A}(g \circ f = \mathsf{id}) \times (f \circ h = \mathsf{id})$$

Ми пишемо $A\mathsf{Equiv}B$ для суми $\sum_{f:A\to B} \mathsf{isEquiv}(f)$.

HoTT є розширенням інтенсіональної теорії типів з ієрархією універсумів, що задовольняють аксіому унівалентності:

$$\mathsf{univ}_i : \textstyle\prod_{A,B:\mathcal{U}_i} \mathsf{isEquiv}(\lambda p.\,(p_!, \cdots) : A =_{\mathcal{U}_i} B \to A\mathsf{Equiv}B)$$

Ми будемо опускати i в univ_i та $\mathcal{U}_i$ і ігнорувати питання розміру, якщо вони не є суттєвими. Унівалентність породжує велику кількість шляхів у $\mathcal{U}$, які відрізняються від refl. Нас часто цікавлять типи, які є тривіальними, мають лише тривіальні шляхи або тривіальні шляхи між шляхами і так далі. Ці умови організовані в сімейство предикатів, що називаються рівнем усічення $(-2, -1, 0, \ldots)$ типу. Ми будемо використовувати лише перші три рівні, стверджуючи, що тип є стягуваним, (гомотопічною) пропозицією або множиною:

$$\mathsf{isContr}(A) = \textstyle\sum_{a:A} \prod_{b:A} a = b \qquad \mathsf{isProp}(A) = \prod_{a,b:A} \mathsf{isContr}(a = b)$$

$$\mathsf{isSet}(A) = \textstyle\prod_{a,b:A} \mathsf{isProp}(a = b)$$

Proposition 11 ([51], див. також [45])**.** *Усі теоретико-типові модельні топоси (і, отже, ∞-топоси Гротендіка) моделюють* HoTT.

Ми також матимемо нагоду використовувати різні *вищі індуктивні типи* (ВІТ). Семантика ВІТ є складною і безпосередньо не розглядається наведеним вище результатом [26]. Зокрема, хоча [51] показує, що вищезгадана модель підтримує всі вищі індуктивні типи, він не показує, що універсуми є строго замкненими щодо цих конструкцій. Хоча робота над отриманням цього результату триває, легко показати, що універсуми є *слабко* замкненими щодо цих конструкцій. Наприклад, існує тип $D : \mathcal{U}_0$, такий що $D \mathsf{Equiv} A \amalg_C B$ щоразу, коли $A, B, C : \mathcal{U}_0$. Відповідно, ми будемо припускати, що наші універсуми замкнені щодо вищих індуктивних типів, хоча і лише з пропозиційними β-правилами.

2.2 Симпліціальна теорія типів

Маючи під рукою HoTT, ми переходимо до симпліціальної теорії типів. Це розширення HoTT кількома аксіомами, які дозволяють нам розглядати (певні) типи як $(\infty, 1)$-категорії, що надалі будуть називатися просто категоріями. Ми, відповідно, опускатимемо префікс $(\infty, 1)$- або ∞- скрізь. По-перше і найфундаментальніше, ми додаємо наступне:

Axiom Б. *Існує множина $\mathbb{I}$, що утворює обмежену дистрибутивну ґратку $(0, 1, \vee, \wedge)$ таку, що $\prod_{i,j:\mathbb{I}} i \leq j \vee j \leq i$ виконується.*

Ми розглядаємо $\mathbb{I}$ як *напрямлений інтервал*, і [46] використовують це, щоб надати кожному типу поняття синтетичного морфізму:

Definition 12. Синтетичний морфізм $f : \mathsf{hom}_X(x, y)$ між $x, y : X$ - це функція $f : \mathbb{I} \to X$ разом з пропозиційними рівностями $f\,0 =_X x$ та $f\,1 =_X y$.

***Remark* 13.** В [46], синтетичний інтервал визначається як більш примітивна судженнява структура і $\mathsf{hom}(x, y)$ використовує строгі типи розширення. Це дає більше визначених рівностей: $f\,0$ та x співпадатимуть за означенням, коли $f : \mathsf{hom}(x, y)$ (і аналогічно для $f\,1$ та y). Проте, підхід суджень не дозволяє прямо включити $\mathbb{I}$ як звичайний тип, і його взаємодії з модальностями (Розділ 2.4) є складними. З цих причин, ми працюємо з простішим, але менш строгим означенням $\mathsf{hom}(x, y)$. В системі, де обидва доступні, ці два поняття еквівалентні[10].

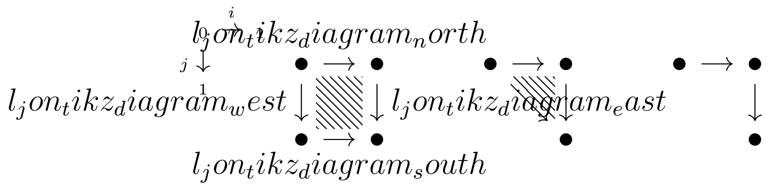

Рис. 2: Visualization of $\mathbb{I}^2$, Δ^2, and Λ^2_1.

Звісно, ми можемо визначити тотожний морфізм $\mathsf{id}x : x \to x$ як $\lambda\_.\,x$. Більше того, кожна функція $f : X \to Y$ автоматично має дію на синтетичні морфізми $\alpha : \mathbb{I} \to X$ шляхом пост-композиції $f \circ \alpha : \mathbb{I} \to Y$. В такому випадку ми часто пишемо $f(\alpha)$.

З $\mathbb{I}$ ми негайно отримуємо n-куби $\mathbb{I}^n$ і з них ми можемо виділити симплекси Δ^n, межі $\partial\Delta^n$ та роги Λ^n_k. Зокрема, $\Delta^2 \to X$ представляє 2-комірку в X, що свідчить про композицію двох стрілок, і $\Lambda^2_1 \to X$ представляє пару композиційних стрілок (без композиції). Ми наводимо означення цих типів нижче:

$$\Delta^n = \{(i_1, \ldots, i_n) : \mathbb{I}^n \mid i_1 \geq i_2 \geq \cdots \geq i_n\}$$

$$\Lambda^2_1 = \{(i, j) : \mathbb{I}^2 \mid i = 1 \vee j = 0\}$$

Notation 14. Ми пишемо $i : \Delta^n$ $(0 \leq i \leq n)$ як скорочений запис для послідовності з i копій 1, за якими слідує 0: $(1, 1, \ldots, 0, \ldots)$.

A map $f : \Delta^2 \to X$ is said to witness that the composite of $f(-, 0)$ followed by $f(1, -)$ is $\lambda i.\, f(i, i)$. We emphasize that this is *data*; there can be many distinct f's witnessing the same composition as X may have many non-equivalent 2-cells with the same boundary. By the same token however, it is not always the case that a pair of composable morphisms $\Lambda^2_1 \to X$ extends to a composition datum $\Delta^2 \to X$. This is precisely because not every type in STT can be regarded as a category; even though we have defined $\mathsf{hom}_X(x, y)$ for every X, there is no *a priori* way of *composing* these morphisms. Precategories are types for which all composites exist:

Definition 15. A *precategory* is a type X satisfying the Segal condition: the inclusion $\Lambda^2_1 \to \Delta^2$ induces an equivalence $\mathsf{isEquiv}(X^{\Delta^2} \to X^{\Lambda^2_1})$.

Roughly, the Segal condition ensures that every pair of composable morphisms in X extends (uniquely) to a 2-cell witnessing their composition and, in particular, there is an induced composition function $\mathsf{hom}(x,y)\times\mathsf{hom}(y,z)\to\mathsf{hom}(x,z)$. Uniqueness automatically ensures that this operation is associative and unital. The definition of a category refines this slightly. In a precategory X we are able to define the type of isomorphisms $x\cong y$ between $x,y:X$ and so there are two potentially distinct types of evidence for x and y being identical: $x=_X y$ and $x\cong_X y$. A category is a precategory for which these two types are canonical equivalent.

Definition 16. $\alpha : \mathsf{hom}(x,y)$ is an isomorphism ($\mathsf{isIso}(\alpha)$) if there exist $\beta_0,\beta_1 : \mathsf{hom}(y,x)$ such that $\beta_0\circ f = \mathsf{id}, f\circ\beta_1 = \mathsf{id}$.[4] We write $\mathsf{Iso}(x,y)$ or $x\cong y$ for the subtype of isomorphisms.

Definition 17. A precategory C is a *category* if it satisfies the Rezk condition:

$$\textstyle\prod_{x,y:C}\mathsf{isEquiv}(\mathsf{idtoiso} : (x=y)\to\mathsf{Iso}(x,y))$$

where $\mathsf{idtoiso}(\mathsf{refl}) := \mathsf{id}$. If every morphism in C is an isomorphism, then C is a *groupoid*.

Example 18. $\mathbb{I},\Delta^n,\mathbb{I}^n$ are all categories [19].

Lemma 19. *C is a groupoid if and only if* $\mathsf{isEquiv}(C^{!_\mathbb{I}} : C\to C^{\mathbb{I}})$ (C *is* $\mathbb{I}$-null [47]).

[46] develop the basic theory of these synthetic categories. As noted above, every function has an action on morphisms and *op. cit.* shows that this action preserves compositions and identities and therefore defines a functor. They also show that $C\to D$ is then a category whenever D is, and that synthetic morphisms $\mathsf{hom}_{D^C}(f,g)$ are precisely natural transformations. One can reformulate various classical categorical notions rather directly:

Definition 20 ([4])**.** A natural transformation $\alpha : \mathsf{hom}_{C^I}(\mathsf{const}(c),F)$ witnesses c as the limit of $F : C^I$ if α induces an equivalence $\mathsf{hom}(c',c)\mathsf{Equiv}\mathsf{hom}(\mathsf{const}(c'),F)$ for all c'.

Definition 21. An adjunction between two categories C,D consists of a pair of functions $f : C\to D$ and $g : D\to C$ with a natural isomorphism $\iota : \prod_{c,d}\mathsf{hom}(f(c),d)\mathsf{Equiv}\mathsf{hom}(c,g(d))$.

While we have given a few examples of categories above, a notable type that is *not* category is the universe $\mathcal{U}$. Maps $A:\mathbb{I}\to\mathcal{U}$ are too unstructured

[4]This is precisely the HoTT equivalence but recast into synthetic morphisms.

to compose and, in particular, correspond neither to functions $A(0) \to A(1)$ nor $A(1) \to A(0)$ (consider $\lambda i.\, i = 0$ or $\lambda i.\, i = 1$). In Section 2.4, we shall discuss the subuniverse $\mathcal{S}$ constructed previously [19], which is a category of groupoids whose morphisms correspond to functions. To properly situate this definition, we recall what it means for $X : A \to \mathcal{U}$ to be covariant [46], giving an assignment from morphisms $\mathsf{hom}(a_0, a_1)$ to functions $X(a_0) \to X(a_1)$.

Notation 22. Given $X : A \to \mathcal{U}$ we write $\widetilde{X}$ for $\sum_{a:A} X(a)$.

Definition 23. A family $X : A \to \mathcal{U}$ is *covariant* if for every $a : \mathsf{hom}(a_0, a_1)$ and $x_0 : X(a_0)$, the following is contractible:

$$\mathsf{Lift}(a, x_0) = \textstyle\sum_{x_1 : X(a_1)} \sum_{x : \mathsf{hom}((a_0, x_0), (a_1, x_1))} \pi_1(x) =_{\mathsf{hom}(a_0, a_1)} a$$

Here, x is a morphism in $\widetilde{X}$. We further say the projection $\sum_{a:A} X(a) \to A$ is covariant when X is. For a general map $\pi : X \to A$ we write X_a for $\sum_{x:X} \pi(x) = a$ and say π is covariant when $\lambda a.\, X_a$ is.

Since $\mathsf{Lift}(a, x_0)$ is contractible it has an inhabitant x_1. This yields a function on $a, x_0 \mapsto x_1$ which defines $a_! : X(a_0) \to X(a_1)$. The contractibility of $\mathsf{Lift}(a, x_0)$ ensures that these functions compose correctly, etc.

Lemma 24. *A family $X : A \to \mathcal{U}$ is covariant if and only if the map $\bar{\pi} := \lambda p.\, (p(0), \pi_1 \circ p) : \widetilde{X}^{\mathbb{I}} \to \widetilde{X} \times_A A^{\mathbb{I}}$ is an equivalence.*

In Sections 4.1 and 5.3, we shall briefly use a weakening of covariance:

Definition 25. A family $X : A \to \mathcal{U}$ is cocartesian if $\widetilde{X}^{\mathbb{I}} \to A^{\mathbb{I}} \times_{A^{\{1\}}} \widetilde{X}^{\{1\}}$ is a right adjoint $\ell \dashv \bar{\pi}$ such that $\bar{\pi} \circ \ell = \mathsf{id}$.

One can give an equivalent characterization in terms of cocartesian morphisms and show that e.g., every morphism in D can be factored as a cocartesian morphism followed by a vertical morphism:

Theorem 26 ([10])**.** *If a map $\pi : D \to C$ is cocartesian then for every $c : \mathbb{I} \to C$, $x_0 : C_{c(0)}$, the category $\mathsf{Lift}(c, x_0)$ has an initial object.*

Dually, one can consider contravariant and cartesian families and fibrations.

Finally, we note that since categories and groupoids are defined by certain *orthogonality* conditions, by [47] they define reflective subuniverses.

Proposition 27. *There are idempotent monads $\bigcirc_{\mathsf{cat}}, \bigcirc_{\mathsf{grpd}}$ such that, e.g., $\bigcirc_{\mathsf{cat}} X$ is a category and $C^{\bigcirc_{\mathsf{cat}} X} \mathsf{Equiv} C^X$ when C is a category.*

Proposition 28 ([46]). *When Theorem 11 is specialized to simplicial spaces* $(\widehat{\Delta})$*, the resulting model validates Axiom B and in this model categories are realized by* ∞*-categories (modeled by complete Segal spaces) and groupoids by* ∞*-groupoids.*

2.3 Modal homotopy type theory

Many theorems in category theory require the ability to quantify over the objects in a category, e.g., "if $\alpha : F \to G$ is a natural transformation of functors $\mathcal{C} \to \mathcal{D}$ and each α_c is invertible, then α is invertible". A version of this is proven by [46]: $\left(\prod_{c:C} \mathsf{isIso}(\lambda i.\, \alpha\, i\, c)\right) \to \mathsf{isIso}(\alpha)$, but this is subtly different as we discuss below. In fact, as it stands we cannot directly capture the classical statement in STT.

To understand the divergence between the STT and classical results, note that by working internally to type theory when proving $\prod_{c:C} \mathsf{isIso}(\lambda i.\, \alpha\, i\, c)$ we cannot assume that c is just an object in C: since it is an arbitrary element, we have to assume it is constructed in an arbitrary context which might contain, e.g., a copy of $\mathbb{I}$ such that c represents a synthetic morphism. In fact, if we unfold the above type into the model we find that constructing $\prod_{c:C} \mathsf{isIso}(\lambda i.\, \alpha\, i\, c)$ already entails proving that the chosen inverses are natural. A great deal of the power of simplicial type theory comes from this implicit naturality, but it makes this particular result weaker. After all, its purpose in standard category theory was that in this particular situation, *a priori* unnatural choices of inverses will automatically be natural. Moreover, we shall encounter theorems that are simply false when naively translated in this way.

Accordingly, to make STT practical we must extend it with *modalities*: unary type constructors distinguished by their failure to respect substitution or apply in arbitrary contexts. For instance, we shall eventually equip STT with a modality $-_\flat$ which discards all non-invertible synthetic morphisms from a type to produce its *core*, which we then use to faithfully encode pointwise invertibility (see Example 31).

A complete reference to the modal type theory we use—MTT [15]—is given by [18] and we record formal rules in Section A. Fortunately, the rules for, e.g., $\sum$-types are unaffected by the addition of modalities. Accordingly, for brevity we only recall the new rules which must be added to MLTT to extend HoTT with modalities à la MTT.

MTT is parameterized by a *mode theory*: a strict 2-category describing the collection of modalities (the morphisms) available along with the natural transformations between them (the 2-cells). We use μ, ν, ξ to range over modalities. In the case of simplicial type theory, our mode theory will have only one object along with a handful of generating modalities and 2-cells. There are four generating modalities $\flat, \sharp, \mathsf{op}, \mathsf{tw}$ subject to the following equations:

$$\flat = \flat \circ \flat = \flat \circ \sharp = \flat \circ \mathsf{op} = \mathsf{tw} \circ \flat \qquad \sharp = \sharp \circ \sharp = \sharp \circ \flat = \sharp \circ \mathsf{op} \qquad \mathsf{op} \circ \mathsf{op} = \mathsf{id}$$

We further require the following generating 2-cells:

$$\epsilon : \flat \to \mathsf{id} \qquad \zeta : \mathsf{id} \to \sharp \qquad \tau : \mathsf{tw} \cong \mathsf{tw} \circ \mathsf{op} \quad (\text{with } \tau^{-1}) \qquad \pi_0^{\mathsf{tw}} : \mathsf{tw} \to \mathsf{op}$$

$$\pi_1^{\mathsf{tw}} : \mathsf{tw} \to \mathsf{id}$$

These 2-cells are likewise subject to a number of equations. For ϵ and ζ, we require $\zeta \cdot \sharp = \sharp \cdot \zeta = \mathsf{id}$ viewing all of these as 2-cells $\sharp \to \sharp$ and $\flat \cdot \zeta = \mathsf{id} : \flat \to \flat$ (using $\cdot$ to denote whiskering). We require the dual equations on ϵ. For the remaining four 2-cells, we require that the following diagrams commute:

$$\begin{tikzcd} & \mathsf{tw} \arrow[dl, "\pi_0^{\mathsf{tw}}"'] \arrow[d, "\tau"] \arrow[dr, "\pi_1^{\mathsf{tw}}"] & \\ \mathsf{op} & \mathsf{tw} \circ \mathsf{op} \arrow[l, "\pi_1^{\mathsf{tw}} \cdot \mathsf{op}"] \arrow[r, "\pi_0^{\mathsf{tw}} \cdot \mathsf{op}"'] & \mathsf{id} \end{tikzcd} \qquad \begin{tikzcd} & \flat \arrow[dl, "\epsilon \cdot \mathsf{op}"'] \arrow[d, "\mathsf{tw} \cdot \epsilon"] \arrow[dr, "\epsilon"] & \\ \mathsf{op} & \mathsf{tw} \arrow[l, "\pi_1^{\mathsf{tw}}"] \arrow[r, "\pi_0^{\mathsf{tw}}"'] & \mathsf{id} \end{tikzcd}$$

Each morphism μ in the mode theory induces a modal type $-_\mu$. We will describe the rules for these modal types in a moment, but first we give some idea of what they are intended to denote. For now this is merely intuition, though the axioms and model described in Section 2.4 will make it so. As already mentioned, $-_\flat$ removes all non-identity synthetic morphisms from a type. $-_\sharp$ is the right adjoint to this operation and so it discards all non-identity morphisms but then freely adds all morphisms so that an n-simplex $\Delta^n \to \mathsf{X}_\sharp$ is exactly a collection of n points in X. [5] Next, $-_{\mathsf{op}}$ sends a type to its opposite and, in particular, reverses the directions of all synthetic

[5]Note that while $\mathsf{X}_\flat$ will always be an ∞-category, in fact an ∞-groupoid, the same is not true of $\mathsf{X}_\sharp$. In particular, $\mathsf{X}_\sharp$ will hardly ever satisfy the Rezk condition.

morphisms. Finally, $-_{\mathsf{tw}}$ sends a type to its corresponding type of *twisted arrows*; we shall analyze it in more depth in Section 3.

The formation rule for $-_\mu$ is complex: the entire point of modalities is that $\Gamma \vdash A$ does not imply $\Gamma \vdash \mathsf{A}$. Instead, MTT introduces a novel form of context operation which acts like a "left adjoint" to $-_\mu$:

$$\frac{\vdash \Gamma}{\vdash \Gamma, \{\mu\}} \qquad \frac{\Gamma, \{\mu\} \vdash A}{\Gamma \vdash \mathsf{A}} \qquad \frac{\Gamma, \{\mu\} \vdash a : A}{\Gamma \vdash \mathsf{mod}_\mu(a) : \mathsf{A}}$$

We refer to $\{\mu\}$ as a *modal restriction*. It is helpful to compare A_μ with dependent products and, therefore, to see $-, \{\mu\}$ as extending the context by something akin to a substructural "μ variable" [6, 7]. The real force of modalities comes through how these $\{\mu\}$s interact with variables. In particular, it is not the case that $\Gamma, x : A, \{\mu\} \vdash x : A$; since $-, \{\mu\}$ is intended to model a left adjoint, we cannot generally assume that there is a weakening substitution $\Gamma, \{\mu\} \to \Gamma$. Instead, we alter the rule extending a context with a variable so that each variable is annotated with a modality:

$$\frac{\vdash \Gamma \qquad \Gamma, \{\mu\} \vdash A}{\vdash \Gamma, x :_\mu A} \qquad \frac{\alpha : \mu \to \mathsf{mods}(\Gamma_1)}{\Gamma_0, x :_\mu A, \Gamma_1 \vdash x^\alpha : A^\alpha}$$

In the above, x^α is the new form of variable rule while A^α is an admissible operation on the syntax which traverses the term A and appropriately updates all free variables y^β occurring within A and modifying β appropriately using α. In particular, if A is closed then $A^\alpha = A$. In the formal syntax, both x^α and A^α are realized by form of substitution, see [16] for further details.

The original context extension is given by taking $\mu = \mathsf{id}$. In the second rule, $\mathsf{mods}(\Gamma_1)$ is the composite $\nu_0 \circ \nu_1 \circ \cdots$ of all the $\{\nu_i\}$s occurring in Γ_1 (and is id if there are no such occurrences). In other words, a variable with annotation μ can be used precisely when it occurs behind a series of modal restrictions for which there is a 2-cell navigating from μ to this composite. It is therefore in the variable rule where the 2-cells comes into play.

Lemma 29. *If* $\Gamma, x :_\mu A \vdash B$, $\Gamma, x :_\mu A \vdash b : B$, *and* $\Gamma, \{\mu\} \vdash a : A$, *then* $\Gamma \vdash B[a/x]$ *and* $\Gamma \vdash b[a/x] : B[a/x]$.

The final piece of the puzzle is the elimination rule for modalities. Roughly, this rule says that modal annotations are equivalent to modal types "from the perspective of a type", i.e., that giving an element in context $\Gamma, x :_\nu \mathsf{A}_\mu$ is the same as giving one in $\Gamma, x :_{\nu\circ\mu} A$. This concretely amounts to the following pattern-matching rule which allows us to assume that $x :_\nu \mathsf{A}_\mu$ is of the form $\mathsf{mod}_\mu(y)$ where $y :_{\nu\circ\mu} A$:

$$\frac{\Gamma, x :_\nu \mathsf{A} \vdash B \qquad \Gamma, y :_{\nu\circ\mu} A \vdash b : B[\mathsf{mod}_\mu(y)/x] \qquad \Gamma, \{\nu\} \vdash a : \mathsf{A}}{\Gamma \vdash \mathsf{let}\ \mathsf{mod}_\mu(y) \leftarrow a\ \mathsf{in}\ b : B[a/x]}$$

$$\frac{\Gamma, x :_\nu \mathsf{A} \vdash B \qquad \Gamma, y :_{\nu\circ\mu} A \vdash b : B[\mathsf{mod}_\mu(y)/x] \qquad \Gamma, \{\nu \circ \mu\} \vdash a : A}{\Gamma \vdash (\mathsf{let}\ \mathsf{mod}_\mu(y) \leftarrow \mathsf{mod}_\mu(a)\ \mathsf{in}\ b) = b[a/y] : B[\mathsf{mod}_\mu(a)/x]}$$

While these rules account for all of the necessary extensions to handle modal types, we avail ourselves of a convenience feature as well, modal $\prod$-types:

$$\frac{\Gamma, x :_\mu A \vdash b : B}{\Gamma \vdash \lambda x.b : \prod_{x:_\mu A} B} \qquad \frac{\Gamma \vdash f : \prod_{x:_\mu A} B \quad \Gamma, \{\mu\} \vdash a : A}{\Gamma \vdash f(a) : B[a/x]}$$

Notation 30. "If $c :_\flat C$, then $\Phi(c)$" signifies $\prod_{c:_\flat C} \Phi(c)$.

Example 31. A faithful translation of "pointwise invertibility implies invertibility" where $C, D :_\flat \mathcal{U}$ and $\alpha : C \times \mathbb{I} \to D$ is $\left(\prod_{c:_\flat C} \mathsf{isIso}(\lambda i.\, \alpha\, i\, c)\right) \to \mathsf{isIso}(\alpha)$

Immediately from these rules, we may prove the following:

Proposition 32 ([15])**.**

- $-_\mu$ *commutes with* $\sum$ *and* 1
- $\mathsf{comp} : -_{\nu\mu}\mathsf{Equiv}-_{\mu\circ\nu}$ *and* $-_{\mathsf{id}}\mathsf{Equivid}$
- *If* $\alpha : \mu \to \nu$*, then there is a map* $\mathsf{coe}^\alpha : -_\mu \to -^\alpha{}_\nu$*.*
- $\mathsf{transp} : \mathsf{A}_\flat \to \mathsf{B}_\flat\mathsf{EquivA} \to \mathsf{B}_{\sharp\flat}$
- $\mathsf{transp} : \mathsf{A}_{\mathsf{op}} \to \mathsf{B}_\flat\mathsf{EquivA} \to \mathsf{B}_{\mathsf{op}\flat}$

The first point yields a function $(\circledast) : \mathsf{A} \to \mathsf{B} \to \mathsf{A} \to \mathsf{B}$*.*

When it will not confusion, we will suppress the equivalences $\mathsf{A}_{\nu\mu}\mathsf{EquivA}_{\mu\circ\nu}$ and $\mathsf{A}_{\mathsf{id}}\mathsf{Equiv}A$. Furthermore, as there is no ambiguity, we suppress ϵ (and its whiskerings) and simply write x instead of x^ϵ and similarly for $\zeta : \mathsf{id} \to \sharp$. Consequently, if $A :_\flat \mathcal{U}$ then we are able to simply write A_{tw} rather than $\mathsf{A}^{\mathsf{tw}\cdot\epsilon}{}_{\mathsf{tw}}$. By convention, we also avoid writing x^{id}.

Notation 33. If $\Gamma, \{\mu\} \vdash f : A \to B$, we write $f^\dagger$ for the function $\mathsf{mod}_\mu(f) \circledast -$.

***Remark* 34.** Since we shall capitalize on the fact repeatedly, we note that coe^α is always suitably natural. For instance, fix $f :_\flat A \to B$. Then we construct a path $\alpha : \mathsf{coe}^{\pi_1^{\mathsf{tw}}} \circ f^\dagger = f \circ \mathsf{coe}^{\pi_1^{\mathsf{tw}}}$ as follows:

$$\alpha = \mathsf{funext}(\lambda x.\, \mathsf{let}\ \mathsf{mod}_{\mathsf{tw}}(x_0) \leftarrow x\ \mathsf{in}\ \mathsf{refl})$$

Since this path is essentially a commuting conversion (it is given by induction to allow $\mathsf{coe}^{\pi_1^{\mathsf{tw}}}$ and $f^\dagger$ to reduce) it is fully coherent, with higher paths being likewise constructed by induction and then reflexivity.

In general, $-_\mu$ need not commute with propositional equality. However, this is true in our intended models and so we impose it as an axiom:

Axiom Β. *The map* $\mathsf{mod}_\mu(a) = \mathsf{mod}_\mu(b) \to \mathsf{a} = \mathsf{b}$ *sending* refl *to* $\mathsf{mod}_\mu(\mathsf{refl})$ *is an equivalence for all* $a, b :_\mu A$.

To be very precise, this map is defined by path induction in the family of types $\lambda(x, y : \mathsf{A}).\, \mathsf{let}\ \mathsf{mod}_\mu(a) \leftarrow x\ \mathsf{in}\ \mathsf{let}\ \mathsf{mod}_\mu(b) \leftarrow y\ \mathsf{in}\ \mathsf{a} = \mathsf{b}$. By [17], there is a computational account of this principle.

Corollary 35. *Each* $-_\mu$ *commutes with pullbacks* $A \times_C B = \sum_{a:A} \sum_{b:B} f(a) =_C g(b)$.

***Remark* 36.** For readers familiar with *spatial type theory* [50], this modal type theory is an extension of spatial type theory to include two additional modalities (op, tw). In particular, the results of [50] that deal with $\flat$ and $\sharp$ can be reproduced in this setting.

2.4 Modalities and simplicial type theory

To connect the modal and simplicial structures, we impose the following axioms motivated by the intended model, as described in Theorem 28 (and more generally $\mathcal{E}^{\Delta^{\mathsf{op}}}$ for an ∞-topos $\mathcal{E}$); see also the work of [33]. First, the opposite map should be an anti-equivalence of $\mathbb{I}$:

Axiom Γ. *There is an equivalence* $\neg : \mathbb{I}_{\mathsf{op}} \to \mathbb{I}$ *which swaps* 0 *for* 1 *and* $\vee$ *for* $\wedge$.

Corollary 37. *We can extend* $\neg$ *to an equivalence* $\neg : \Delta^{\mathsf{n}}{}_{\mathsf{op}}\mathsf{Equiv}\Delta^n$.

Next, we require the two possible notions of discreteness (being $\mathbb{I}$-null or $\flat$-modal) to coincide:

Axiom Д. *If $A :_\flat \mathcal{U}$, then $\mathsf{A}_\flat \to A$ is an equivalence (A is* discrete*) if and only if $A \to A^{\mathbb{I}}$ is an equivalence (A is* $\mathbb{I}$-null*).*

Axiom E. *The canonical map $\mathsf{Bool} \to \mathbb{I}$ is injective and induces an equivalence $\mathsf{BoolEquiv}\mathbb{I}_\flat$.*

Motivated by our intended class of models, we insist that equivalences are jointly detected by Δ^n:

Axiom Ж. *$f :_\flat A \to B$ is an equivalence if and only if the following holds:*

$$\prod_{n:_\flat \mathsf{Nat}} \mathsf{isEquiv}((f_*)^\dagger : \mathsf{\Delta^n} \to \mathsf{A}_\flat \to \mathsf{\Delta^n} \to \mathsf{B}_\flat)$$

Note that since there is a section-retraction pair $\Delta^n \to \mathbb{I}^n \to \Delta^n$, we can replace Δ^n with $\mathbb{I}^n$ in the above principle.

One useful application is the following:

Lemma 38. *A map $\pi :_\flat X \to A$ is covariant if and only if the map $\mathsf{X^{\Delta^n}}_\flat \to \mathsf{X}_\flat \times_{\mathsf{A}_\flat} \mathsf{A^{\Delta^n}}_\flat$ induced by $(0^*)^\dagger$ and $(\pi_*)^\dagger$ is an equivalence for all $n :_\flat \mathsf{Nat}$.*

The axiom for $-_{\mathsf{tw}}$

Finally, we add a new axiom to STT that governs tw. For motivation, we recall some facts about the external definition of the twisted arrow functor $\widehat{\Delta} \to \widehat{\Delta}$ which $-_{\mathsf{tw}}$ is intended to internalize. Classically, $\mathsf{Tw} : \widehat{\Delta} \to \widehat{\Delta}$ is defined as follows:

$$\mathsf{Tw}(X)([n]) = X([n]^{\mathsf{op}} * [n])$$

Here we have written $[n]^{\mathsf{op}} * [n]$ instead of the equivalent $[2n+1]$ to clarify the action of this functor on morphisms: $f \mapsto f^{\mathsf{op}} * f$. (I.e., this corresponds to the join operation on finite linear orders.)

As this functor is defined by precomposition, it is a right adjoint whose left adjoint is defined by left Kan extension. In particular, it sends $\Delta^n : \widehat{\Delta}$ to Δ^{2n+1} (again, with the functorial action given by twisting). As such, there is a universal map $\eta_n : \Delta^n \longrightarrow \mathsf{Tw}(\Delta^{2n+1})$ which, when unfolded, is given by the identity $[2n+1] \longrightarrow [2n+1]$. Universality of η_n amounts to the requirement that each morphism $f : \Delta^n \longrightarrow \mathsf{Tw}(C)$ factors as $\mathsf{Tw}(\hat{f}) \circ \eta_n$ for some unique $\hat{f} : \Delta^{2n+1} \longrightarrow C$. Our axiom governing $-_{\mathsf{tw}}$ axiomatizes this η_n along with the property that $\mathsf{Tw}(-) \circ \eta_n : \mathsf{hom}(\Delta^{2n+1}, C) \to \mathsf{hom}(\Delta^n, \mathsf{Tw}(C))$ is an equivalence. With this external motivation to hand, we proceed to fix some notation and state the axiom which governs $-_{\mathsf{tw}}$.

Notation 39. If $n :_\flat \mathsf{Nat}$, we have canonical maps $i_l : \Delta^n \to \Delta^{2n+1}$, $i_r : \Delta^n \to \Delta^{2n+1}$, and $i_m : \Delta^1 \to \Delta^{2n+1}$ which picks out $\{0, \ldots, n\}$, $\{n+1, \ldots, 2n+1\}$, and $\{n, n+1\}$ respectively. For convenience, we write $\bar{i}_l = i_l^\dagger \circ \neg : \Delta^n \to \mathsf{\Delta^{2n+1}}_{\mathsf{op}}$.

Finally, in the statement of new axiom, we require a procedure which extends a map $f : \Delta^n \to \Delta^m$ to a map $\Delta^{2n+1} \to \Delta^{2m+1}$ which acts appropriately on the images of two inclusions $i_l, i_r : \Delta^n \to \Delta^{2n+1}$. To justify this formally, we introduce the *blunt join* $X \diamond Y$:

$$X \diamond Y = X \amalg_{X \times \{0\} \times Y} (X \times \mathbb{I} \times Y) \amalg_{X \times \{1\} \times Y} Y$$

This is the directed version of the join $X \star Y$ [53, Ch 6] such that $X \diamond Y$ is roughly $X \coprod Y$ with morphisms adjoined to connect each $x : X$ to each $y : Y$.

Lemma 40. *If C is a category, then $C^{\Delta^{m+1+n}} \mathsf{Equiv} C^{\Delta^m \diamond \Delta^n}$.*

Definition 41. If $f :_\flat \Delta^n \to \Delta^m$ and we take $\mathsf{twist}(f) : \Delta^{2n+1} \to \Delta^{2m+1}$ to be the map given given by uniquely extending the map $f^\dagger \diamond f : \mathsf{\Delta^n}_{\mathsf{op}} \diamond \Delta^n \to \mathsf{\Delta^m}_{\mathsf{op}} \diamond \Delta^m$ along the categorical equivalences $\mathsf{\Delta^i}_{\mathsf{op}} \diamond \Delta^i \to \Delta^{2i+1}$.

Axiom И. *For each $n :_\flat \mathsf{Nat}$, there is a (necessarily unique) function $\eta_n :_\flat \Delta^n \to \mathsf{\Delta^{2n+1}}_{\mathsf{tw}}$ such that the following map is an equivalence, for each category $C :_\flat \mathcal{U}$:*

$$\iota :\!- \lambda \mathsf{mod}_\flat(f).\, \mathsf{mod}_\flat(f^\dagger \circ \eta_n) : \mathsf{\Delta^{2n+1}} \to \mathsf{C}_\flat \to \mathsf{\Delta^n} \to \mathsf{C}_{\mathsf{tw}\flat}$$

Additionally, we require that $\tau = (\mathsf{coe}^\neg)^\dagger : \mathsf{\Delta^n}_{\mathsf{tw}} \to \mathsf{\Delta^n}_{\mathsf{op}_{\mathsf{tw}}}$ and that the diagrams in Figure 3 commute (these are mere properties—all objects are sets since $-_\mu$ preserves h-level).

One may visualize ι as ensuring that $\mathsf{\Delta^n} \to \mathsf{C}_{\mathsf{tw}\flat}$ is isomorphic to a $2n+1$ simplex in C:

$$\begin{tikzcd}
c_n \arrow[d] & c_{n-1} \arrow[l] & \cdots \arrow[l] & c_0 \arrow[l] \\
c_{n+1} \arrow[r] & c_{n+2} \arrow[r] & \cdots \arrow[r] & c_{2n}
\end{tikzcd}$$

Under this correspondence, η is the unique map $\Delta^n \to \mathsf{\Delta^{2n+1}}_{\mathsf{tw}}$ given by the identity $\mathsf{id} : \Delta^{2n+1} \to \Delta^{2n+1}$ and is thus the universal n-simplex. The

$$
\begin{tikzcd}
 & \mathsf{\Delta^{2n+1}}_{\mathsf{op}} \\
\Delta^n \arrow[ur, "\bar{i}_l"] \arrow[r, "\eta_n"] \arrow[dr, "i_r"'] & \mathsf{\Delta^{2n+1}}_{\mathsf{tw}} \arrow[u, "\mathsf{coe}^{\pi_0^{\mathsf{tw}}}"'] \arrow[d, "\mathsf{coe}^{\pi_1^{\mathsf{tw}}}"] \\
 & \Delta^{2n+1}
\end{tikzcd}
\qquad
\begin{tikzcd}
\underset{}{\overset{l_j on_t ikz_d iagram_n orth}{\Delta^n}} \arrow[r] \arrow[d, "l_j on_t ikz_d iagram_w est"'] & \mathsf{\Delta^{2n+1}}_{\mathsf{tw}} \arrow[d, "l_j on_t ikz_d iagram_e ast"] \\
\underset{l_j on_t ikz_d iagram_s outh}{\Delta^m} \arrow[r] & \mathsf{\Delta^{2m+1}}_{\mathsf{tw}}
\end{tikzcd}
$$

Рис. 3: Laws for Axiom И.

map π_1^{tw} picks out the bottom row and π_0^{tw} selects the top but *twisted* so that it lands in $\mathsf{C_{op}}$ rather than C. This axiom will only be used in the proof of Theorem 54, where we use $-_{\mathsf{tw}}$ to construct a bifunctorial version of hom.

Proposition 42 ([19])**.** *The model constructed in Theorem 28 extends to a model of modal* HoTT *validating our axioms.*

***Remark* 43.** While our previous work [19] did not handle $-_{\mathsf{tw}}$, the methods employed there scale directly to this situation. In particular, [32] give an explicit description of the necessary twisted arrow operation and show it is a Quillen right adjoint as required to extend the model.

With modalities to hand, a number of results from classical category theory can be proven directly. For instance, the so-called fundamental theorem of ∞-category theory:

Theorem 44. *If $C, D :_\flat \mathcal{U}$ are categories, then $F :_\flat C \to D$ is an equivalence if (1) the induced map $\mathsf{C}_\flat \to \mathsf{D}_\flat$ is surjective, and (2) for any $c, c' :_\flat C$ the map $\mathsf{hom}(c, c') \to \mathsf{hom}(F(c), F(c'))$ is an equivalence.*

Доведення. Suppose (1) and (2) holds. We prove that F is an equivalence using Axiom Ж and fix $n :_\flat \mathsf{Nat}$ such that it suffices to show $\mathsf{isEquiv}(F_*^\dagger : \mathsf{\Delta^n} \to \mathsf{C}_\flat \to \mathsf{\Delta^n} \to \mathsf{D}_\flat)$.

If $n = 0$, then by (1) $F_*^\dagger$ is surjective and by (2) combined with the Rezk condition, it is an embedding. Accordingly, $F_*^\dagger$ is an equivalence in this case. The case for $n = 1$ is an immediate consequence of the cases for $n = 0$ along with (2). In general, since $C^{\Delta^n}\mathsf{Equiv}C^{\Delta^1} \times_C \cdots \times_C C^{\Delta^1}$ by the Segal condition and likewise for D, and $-_\flat$ commutes with pullbacks, the case for $n \geq 2$ follows from $n = 0, 1$. $\square$

2.5 Basic building blocks for categories

Finally, we recall two results from our earlier work [19] that will be used repeatedly within this work to construct new categories. The first is a construction of *full subcategories* using $\sharp$:

Proposition 45. *If $C :_\flat \mathcal{U}$ is a category and $\phi :_\flat \mathsf{C}_\flat \to \mathsf{HProp}$ is a predicate, then*

1. *$C_\phi = \sum_{c:C} \phi(\mathsf{mod}_\flat(c))_\sharp$ is a category,*
2. *the projection map $C_\phi \to C$ induces an equivalence on hom-types,*
3. *$\mathsf{C}_{\phi_\flat} \mathsf{Equiv} \sum_{c:\mathsf{C}_\flat} \phi(c)$, and*
4. *a map $F :_\flat D \to C$ factors through C_ϕ if and only if $\phi(\mathsf{mod}_\flat(F(d)))$ holds for all $d :_\flat D$.*

Corollary 46. *If $C, D :_\flat \mathcal{U}$ are categories and $F :_\flat C \to D$, then the canonical map $\mathsf{hom}(c, c') \to \mathsf{hom}(F(c), F(c'))$ is an equivalence for all $c, c' : C$ (notice the lack of $\flat$!) if and only if it is an equivalence when $c, c' :_\flat C$.*

Next, we recall the construction of the category of groupoids which plays the role of the category of sets in simplicial type theory, e.g., we shall use this category to define presheaves:

Proposition 47. *There is a category $\mathcal{S}_i :_\flat \mathcal{U}_{i+1}$ with an embedding $\mathcal{S}_i \to \mathcal{U}_i$ such that:*

- *If $X : A \to \mathcal{S}_i$, then the composite $A \to \mathcal{U}_i$ is covariant.*
- *The converse holds for $A :_\flat \mathcal{U}_i$, $X :_\flat A \to \mathcal{U}_i$: if X is covariant, then X factors through $\mathcal{S}_i$.*

Corollary 48 (Directed univalence)**.** *$\mathcal{S}^{\mathbb{I}} \mathsf{Equiv} \sum_{X_0, X_1 : \mathcal{S}} X_1^{X_0}$ and composition in $\mathcal{S}$ is the composition of functions.*

Corollary 49. *If $X :_\flat \mathcal{U}$ is a groupoid, then $X : \mathcal{S}$.*

***Remark* 50.** We proved [19] Proposition 47 in a richer variation of STT (*triangulated type theory*). Since we only require the result here, we take it as an "axiom" of sorts to work in a simpler type theory and note that one could extend STT to triangulated type theory to prove this theorem outright.

3 The Yoneda embedding

Within this section, we fix a category $C :_\flat \mathcal{U}$. Our goal is to study the type $\widehat{C} = \mathcal{S}^{\mathsf{C_{op}}}$ of presheaves on C. As $\mathcal{S}$ is a category, so is $\widehat{C}$, and by directed univalence:

Lemma 51. *If* $F, G : \widehat{C}$ *then* $\mathsf{hom}(F, G)\mathsf{Equiv} \prod_{c:\mathsf{C_{op}}} F\,c \to G\,c$

***Remark* 52.** Just as with e.g., completeness, $\widehat{C}$ implicitly fixes a universe level such that $\widehat{C} = \mathsf{C_{op}} \to \mathcal{S}_i$. We may regard i as a parameter or simply take $i = 0$. Occasionally, we shall need to insist that $C\mathsf{Equiv}C'$ where $C' : \mathcal{U}_i$ and in such situations we shall say that C is *small*. We assume all categories are locally small—that each $\mathsf{hom}_C(c, c')$ is small.

One may recast the *fibrational* Yoneda lemma proven by [46] to take advantage of $\widehat{C}$ rather than quantifying over contravariant families as in *op. cit.*:

Lemma 53. *If* $F : \widehat{C}$ *and* $c : \mathsf{C_{op}}$ *then* $F(c) \cong \prod_{c':\mathsf{C_{op}}} \mathsf{hom}_{\mathsf{C_{op}}}(c, c') \to F(c')$

3.1 The twisted arrow category and the Yoneda embedding

In light of this last result, the natural next step is to define a map $C \to \widehat{C}$ which sends $c : C$ to something like $\mathsf{hom}(-, c)$.[6] However, caution is required: $\mathsf{hom}(-, c)$ has type $C \to \mathcal{U}$ and not the required $\mathsf{C_{op}} \to \mathcal{S}$. Upon reflection, the reader should find it surprising that $\mathsf{hom}(-, -) : C \times C \to \mathcal{U}$ at all; if all maps are functorial in STT how can $\mathsf{hom}(-, -)$ be covariant in both arguments? In fact, this is a consequence of the strange behavior of synthetic morphisms in $\mathcal{U}$. While $\mathsf{hom}(-, -)$ is functorial in both arguments, the lack

[6]Here we see why C must be flat: we wish to discuss both C and $\mathsf{C_{op}}$. It is helpful to understand $C :_\flat \mathcal{U}$ as a *closed* type which depends on nothing in the context and, in particular, need not be treated functorially.

of directed univalence for $\mathcal{U}$ makes this useless. This strangeness ensures that $\mathsf{hom}(-,-)$ does not restrict to a function into $\mathcal{S}$.

What is required instead is a function $\Phi : \mathsf{C_{op}} \times C \to \mathcal{S}$ such that $\Phi(\mathsf{mod_{op}}(c), -) = \mathsf{hom}(c, -)$ whenever $c :_\flat C$, i.e., a function that agrees on objects with $\mathsf{hom}(-,-)$ and has the same functoriality in the second argument, but takes $\mathsf{C_{op}}$ as its first argument. In fact, it is highly non-obvious where such a function should come from; [49, p. xii] specifically highlight this construction as remarkably subtle in ∞-category theory. It is for this reason that we introduced $-_{\mathsf{tw}}$. Recall the visualization of $\Delta^{\mathsf{n}} \to \mathsf{C_{tw\flat}}$:

$$\begin{tikzcd}
c_n \arrow[d] & c_{n-1} \arrow[l] & \cdots \arrow[l] & c_0 \arrow[l] \\
c_{n+1} \arrow[r] & c_{n+2} \arrow[r] & \cdots \arrow[r] & c_{2n}
\end{tikzcd} \tag{1}$$

The projection to $\mathsf{C_{op}}$ gives the top row and the map to C yields the bottom. This visualization for n-simplices is very similar to that of $C^{\mathbb{I}} = \sum_{c_0,c_1} \mathsf{hom}(c_0, c_1)$, but the top row has been twisted to ensure that one restriction lands in $\mathsf{C_{op}}$ as required for a bifunctorial version of $\mathsf{hom}(-,-)$:

Theorem 54. *If $C :_\flat \mathcal{U}$ is a category, then the following holds:*

- *The map $\langle \mathsf{coe}^{\pi_0^{\mathsf{tw}}}, \mathsf{coe}^{\pi_1^{\mathsf{tw}}} \rangle : \mathsf{C_{tw}} \to \mathsf{C_{op}} \times C$ straightens to $\Phi : \mathsf{C_{tw}} \times C \to \mathcal{S}$.*
- *For every $c :_\flat C$, the map $\alpha_c : \mathsf{hom}(\mathsf{hom}(c,-), \Phi(\mathsf{mod_{op}}(c), -))$ induced by the Yoneda lemma (Lemma 53) applied to $\iota(\mathsf{mod}_\flat(\mathsf{id}c)) : \Phi(\mathsf{mod_{op}}(c), c)$ is an equivalence.*

Lemma 55. *Given $f :_\flat \Delta^1 \to C$ and let $\bar{f} = \iota(\mathsf{mod}_\flat(f)) : \mathsf{C_{tw}}$, then there exist paths:*

$$\theta(f)_0 : (\mathsf{coe}^{\pi_0^{\mathsf{tw}}} \circ \mathsf{extract}(\bar{f}))(*) = \mathsf{mod_{op}}(f(0)) \quad \theta(f)_1 : (\mathsf{coe}^{\pi_1^{\mathsf{tw}}} \circ \mathsf{extract}(\bar{f}))(*) = f(1)$$

These paths are natural in C so that e.g., the two paths of the following shape induced by $\theta(g \circ f)_1$ and $\theta(f)_1$ with the naturality of $\mathsf{coe}^{\pi_1^{\mathsf{tw}}}$ at g agree:

$$(\mathsf{coe}^{\pi_1^{\mathsf{tw}}} \circ \mathsf{extract}(\iota(\mathsf{mod}_\flat(g \circ f))))(*) = g(f(1))$$

Here $* : \Delta^0$ is the unique element of the unit type.

Доведення. We show the second, as they are symmetric. We define θ_1 using the naturality of $\mathsf{coe}^{\pi_1^{\mathsf{tw}}}$ and the behavior of $\mathsf{coe}^{\pi_1^{\mathsf{tw}}}$ on η from Figure 3:

$$(\mathsf{coe}^{\pi_1^{\mathsf{tw}}} \circ \mathsf{extract}(\iota(\mathsf{mod}_\flat(f))))(*)$$

$$
\begin{aligned}
&= (\mathsf{coe}^{\pi_1^{\mathsf{tw}}} \circ f^{\dagger} \circ \eta_0)(*) && \\
&= (f \circ \mathsf{coe}^{\pi_1^{\mathsf{tw}}} \circ \eta_0)(*) && \text{By naturality, Remark 34} \\
&= f(1) && \text{By the first diagram in Figure 3}
\end{aligned}
$$

To prove that θ_1 is natural in C, we observe that the terms agree up to a commuting conversion of elimination rules for modal types. Accordingly, we may prove that these two paths agree by induction on $\eta_0(*)$ and then reflexivity. □

Proof of the Theorem 54. We begin by showing that the map $\langle \mathsf{coe}^{\pi_0^{\mathsf{tw}}}, \mathsf{coe}^{\pi_1^{\mathsf{tw}}} \rangle : \mathsf{C_{tw}} \to \mathsf{C_{op}} \times C$ is a covariant family. By Lemma 38 the following map induced by $\{0\} : \Delta^0 \to \Delta^n$ is an equivalence:

$$\epsilon : \mathsf{C}^{\Delta^{n}}_{\mathsf{tw}\,\flat} \to \left(\mathsf{C}_{\mathsf{tw}\flat} \times_{\mathsf{C}_{\mathsf{op}_\flat} \times \mathsf{C}_\flat} (\mathsf{C}^{\Delta^{n}}_{\mathsf{op}\,\flat} \times \mathsf{C}^{\Delta^{n}}{}_\flat)\right)$$

For convenience, we begin by applying a few modal transformations (in particular, using $\mathsf{transp} : \mathsf{A} \to \mathsf{B}_{\mathsf{op}_\flat}\mathsf{Equiv}\mathsf{A}_{\mathsf{op}} \to \mathsf{B}_\flat$) such that it suffices to show that the following map is an equivalence:

$$\epsilon' : \mathsf{C}^{\Delta^{n}}_{\mathsf{tw}\,\flat} \to \left(\mathsf{C}_{\mathsf{tw}\flat} \times_{\mathsf{C}_\flat \times \mathsf{C}_\flat} (\mathsf{C}^{\Delta^{n}{}^{\mathsf{op}}}{}_\flat \times \mathsf{C}^{\Delta^{n}}{}_\flat)\right)$$

To prove this, we shall construct a commutative diagram:

$$
\begin{tikzcd}
\mathsf{C}^{\Delta^{2n+1}}{}_\flat \arrow[r, "l_jon_tikz_diagram_north"] \arrow[d, "l_jon_tikz_diagram_west"'] & \Delta^1 \to \mathsf{C}_\flat \times_{\mathsf{C}_\flat \times \mathsf{C}_\flat} (\mathsf{C}^{\Delta^{n}{}^{\mathsf{op}}}{}_\flat \times \mathsf{C}^{\Delta^{n}}{}_\flat) \arrow[d, "l_jon_tikz_diagram_east"] \\
\mathsf{C}^{\Delta^{n}}_{\mathsf{tw}\,\flat} \arrow[r, "l_jon_tikz_diagram_south"'] & \mathsf{C}_{\mathsf{tw}\flat} \times_{\mathsf{C}_\flat \times \mathsf{C}_\flat} (\mathsf{C}^{\Delta^{n}{}^{\mathsf{op}}}{}_\flat \times \mathsf{C}^{\Delta^{n}}{}_\flat)
\end{tikzcd}
\tag{2}
$$

We define ϕ momentarily, but we first remark that (ι, id) is well-formed because of Lemma 55, which ensures that applying ι and then evaluating commutes appropriately with projection.

Note also that the two vertical maps are equivalences because ι is an equivalence. Accordingly, by 3-for-2, to show ϵ' is an equivalence, it suffices to ensure that ϕ is an equivalence making the diagram commute.[7] We now

[7]We emphasize that the filler for this square is irrelevant. Any filler suffices to show that ϵ' is an equivalence, which in turn implies that ϵ is an equivalence as required.

define ϕ as follows:

$$\phi(\mathsf{mod}_\flat(f)) :\!- (\mathsf{mod}_\flat(f|_{n\le n+1}), (\mathsf{mod}_\flat(f|_{0\le\cdots\le n} \circ \neg), \mathsf{mod}_\flat(f|_{n+1\le\cdots\le 2n+1})), \mathsf{refl})$$

ϕ is given by restricting along a categorical equivalence $(\Delta^{\mathsf{n}}{}_{\mathsf{op}} \diamond \Delta^n \to \Delta^{2n+1})$, so it is an equivalence.

Next, we note that all four of the maps in this diagram are weakly natural in C. For the bottom and top maps, this is an easy observation—the top is given by restriction and the bottom uses restriction along with $\mathsf{coe}^{\pi_1^{\mathsf{tw}}}$, which is also natural by Remark 34. For the left-hand map, this is a consequence of the naturality of ι. For the right-hand map, the only wrinkle is the paths used to witness that ι commutes with evaluating on projections. This requires a filler for a certain path, but this is precisely the naturality coherence supplied by the latter part of Lemma 55.

Finally, we argue that the diagram commutes. Fix $\mathsf{mod}_\flat(f) : \Delta^{\mathsf{2n+1}} \to \mathsf{C}_\flat$. We wish to show that $\epsilon'(\iota(\mathsf{mod}_\flat(f))) = (\iota, \mathsf{id})(\phi(\mathsf{mod}_\flat(f)))$. By naturality, however, we may use f to reduce to the case where $C = \Delta^{2n+1}$ and $f = \mathsf{id}$. In this case, everything involved is a set and so it suffices to argue the diagram commutes when we replace each pullback with a simple product. With this in place, we now calculate:

$$\begin{aligned}
\epsilon'(\iota(\mathsf{mod}_\flat(\mathsf{id}))) &= \epsilon'(\mathsf{mod}_\flat(\eta_n)) \\
&= (\mathsf{mod}_\flat(\eta_n\, 0), (\mathsf{transp}(\mathsf{mod}_\flat(\mathsf{coe}^{\pi_0^{\mathsf{tw}}} \circ \eta_n)), \mathsf{mod}_\flat(\mathsf{coe}^{\pi_1^{\mathsf{tw}}} \circ \eta_n))) \\
&= (\mathsf{mod}_\flat(\eta_n\, 0), (\mathsf{transp}(\mathsf{mod}_\flat(\bar{i}_l)), \mathsf{mod}_\flat(i_r))) \\
&= (\mathsf{mod}_\flat(i_m(\eta_0 *)), (\mathsf{mod}_\flat(i_l \circ \neg), \mathsf{mod}_\flat(i_r))) \\
&= (\iota, \mathsf{id})(\mathsf{mod}_\flat(i_m), (\mathsf{mod}_\flat(i_l \circ \neg), \mathsf{mod}_\flat(i_r))) \\
&= (\iota, \mathsf{id})(\phi(\mathsf{mod}_\flat(\mathsf{id})))
\end{aligned}$$

This completes the first step of the argument. The second is to show that for each $c :_\flat C$ the map $\alpha_c : \mathsf{hom}_{C\to\mathcal{S}}(\mathsf{hom}(c,-), \Phi(\mathsf{mod}_{\mathsf{op}}(c), -))$ is an isomorphism. Passing to total spaces, it suffices to show the following map is an equivalence:

$$\tilde{\alpha}_c = \lambda(d, f).\,(d, f_*(\iota(\mathsf{mod}_\flat(\mathsf{id}c)))) : \textstyle\sum_{d:C} \mathsf{hom}(c, d) \to \sum_{d:C} \Phi(\mathsf{mod}_{\mathsf{op}}(c), d)$$

In the above, f_* is the covariant transport operation on $\Phi(\mathsf{mod}_{\mathsf{op}}(c), -)$. Since both sides of this map are categories, it suffices to show that this map is fully faithful and essentially surjective.

In fact, $\tilde{\alpha}_c$ is an equivalence on objects. To this end, we observe that if $f :_\flat \mathsf{hom}(c, d)$ for some $d :_\flat C$, then we can construct the transport f_* alternatively as follows. Define a path $h : \Delta^1 \to \mathsf{C}_{\mathsf{tw}}$ by $h :\equiv \iota(\mathsf{mod}_\flat(\lambda\_, \_, k.\, f(k)))$, i.e., h corresponds to the following doubly degenerate 3-simplex in C:

$$\begin{tikzcd} c \arrow[d, "\mathsf{id}c"'] & c \arrow[l, "\mathsf{id}c"'] \\ c \arrow[r, "f"'] & d \end{tikzcd}$$

We then consider the morphism $\lambda i.\, (\pi_1^{\mathsf{tw}}(h\, i), h\, i)$ in $\sum_{d:C} \Phi(\mathsf{mod}_{\mathsf{op}}(c), d)$. Using the definition of ι and the naturality of η, this is a morphism from $(c, \iota(\mathsf{mod}_\flat(\mathsf{id}c)))$ to $(d, \iota(\mathsf{mod}_\flat(f)))$. Moreover, the naturality of $\mathsf{coe}^{\pi_1^{\mathsf{tw}}}$ ensures that it lies over f in C. Consequently, $\tilde{\alpha}_c(d, f) = (d, \iota(\mathsf{mod}_\flat(f)))$ when restricted to $d :_\flat C$ and $f :_\flat \mathsf{hom}(c, d)$, which is an equivalence because ι is invertible.

For fully-faithfulness, it suffices to show that the following map is an equivalence:

$$\tilde{\alpha}_c : \mathbb{I} \to \textstyle\sum_{\mathsf{d:C}} \mathsf{hom}(c, d)_\flat \to \mathbb{I} \to \textstyle\sum_{\mathsf{d:C}} \Phi(\mathsf{mod}_{\mathsf{op}}(c), d)_\flat$$

However, as both sides are the total spaces of covariant families, it suffices to show that the following map is an equivalence:

$$\tilde{\alpha}_c : \textstyle\sum_{\mathsf{d}:\mathbb{I}\to\mathsf{C}} \mathsf{hom}(c, d\,0)_\flat \to \textstyle\sum_{\mathsf{d}:\mathbb{I}\to\mathsf{C}} \Phi(\mathsf{mod}_{\mathsf{op}}(c), d\,0)_\flat$$

The conclusion now follows from the previous case. □

Notation 56. We write $\Phi_D : \mathsf{D}_{\mathsf{op}} \times D \to \mathcal{S}$ for the same construction applied to some category D. Within this section, we continue to write Φ as shorthand for Φ_C.

Corollary 57. *If* $c_0 : \mathsf{C}_{\mathsf{op}}$ *and* $c_1 : C$, *then* $\Phi(c_0, c_1) = \Phi_{\mathsf{C}_{\mathsf{op}}}(\mathsf{mod}_{\mathsf{op}}(\mathsf{mod}_{\mathsf{op}}(c_1)), c_0)$.

Доведення. Passing to total spaces, it suffices to find an equivalence $\mathsf{C}_{\mathsf{tw}} \to \mathsf{C}_{\mathsf{op}_{\mathsf{tw}}}$ fitting into the following diagram:

$$\begin{tikzcd} \mathsf{C}_{\mathsf{tw}} \arrow[rr] \arrow[dr] & & \mathsf{C}_{\mathsf{op}_{\mathsf{tw}}} \arrow[dl] \\ & \mathsf{C}_{\mathsf{op}} \times C & \end{tikzcd}$$

The map coe^{τ} precisely satisfies this role: it is invertible because the 2-cell τ is an isomorphism and it fits into the commuting diagram because of the corresponding diagram in the mode theory. $\square$

3.2 The Yoneda lemma

With a bi-functorial version of $\mathsf{hom}(-,-)$ to hand, we can now straightforwardly define the Yoneda embedding Yo and leverage Lemma 53 into a result about Yo:

Definition 58 (Yoneda)**.** $\mathsf{Yo} = \lambda c.\, \Phi(-, c) : C \to \widehat{C}$.

Lemma 59. $\mathsf{hom}(\mathsf{Yo} c, X) \cong X(\mathsf{mod}_{\mathsf{op}}(c))$ *for all* $X : \widehat{C}$ *and* $c :_\flat C$.

Доведення. Since c is $\flat$-annotated, using Theorem 54 and Corollary 57 we have the following identification $\mathsf{hom}_{\mathsf{C_{op}}}(\mathsf{mod}_{\mathsf{op}}(c), -) = \Phi(-, c)$. Moreover, by Lemma 51 we additionally have the following:

$$\textstyle\prod_{c' : \mathsf{C_{op}}} \Phi(c', c) \to X(c') \cong \mathsf{hom}(\mathsf{Yo} c, X)$$

The conclusion now follows by Lemma 53. $\square$

A great deal of category theory is contained within Lemma 59. It shows that Yo is fully-faithful on $\flat$-annotated elements of C and that C is a full subcategory of $\widehat{C}$:

Lemma 60. $\mathsf{Yo} : C \to \widehat{C}$ *induces an equivalence* $C \mathsf{Equiv} \widehat{C}_{\mathsf{isRepr}}$ *where* $\mathsf{isRepr} = \lambda X.\, \sum_{c:C} X = \mathsf{Yo} c$.[8]

While Lemma 59 follows directly from Lemma 53, the above consequence can only be expressed once there exists a *category* of presheaves—something missing from [46]. This opens up a new proof strategy: to prove a result of C, we first prove that it holds for $\mathcal{S}$, then $\widehat{C}$, then that it restricts to the full subcategory. For instance, we may prove the aforementioned characterization of natural isomorphisms:

Theorem 61. *If* $C, D :_\flat \mathcal{U}$ *are categories,* $F, G :_\flat C \to D$*, and* $\alpha :_\flat \mathsf{hom}(F, G)$*, then* $\prod_{c:C} \mathsf{isIso}(\alpha\, c)$ *if* $\prod_{c:_\flat C} \mathsf{isIso}(\alpha\, c)$*.*

[8]Note that $\mathsf{isRepr}(X)$ is a proposition due to Lemma 59 and Corollary 46.

Доведення. Note that this theorem is trivial for $C = \Delta^0$ and for $C = \Delta^1, D = \mathcal{S}$ it is a consequence of Corollary 48. The Segal condition for $\mathcal{S}$ then implies the theorem for $C = \Delta^n, D = \mathcal{S}$.

By Lemma 60, it suffices to assume $D = \widehat{D_0}$. By Axiom Д and Theorem 32 it suffices to show that $\left(\sum_{c:C} \mathsf{isIso}(\alpha\, c)\right) \to \left(\sum_{c:C} \mathsf{isIso}(\alpha\, c)_\sharp\right)$ is an equivalence. By Axiom Ж, it suffices to prove for all n:

$$\mathsf{isEquiv}\Big(\left(\textstyle\sum_{c:C} \mathsf{isIso}(\alpha\, c)\right)^{\Delta^n}{}_\flat \to \left(\textstyle\sum_{c:C} \mathsf{isIso}(\alpha\, c)_\sharp\right)^{\Delta^n}{}_\flat\Big)$$

Unfolding and commuting $\flat$ with $\sum$, it suffices to show that for every $c :_\flat \Delta^n \to C$ the following holds:

$$\textstyle\sum_{\sigma:\Delta^n} \mathsf{isIso}(\alpha(c\,\sigma))\mathsf{Equiv}\sum_{\sigma:_\flat\Delta^n} \mathsf{isIso}(\alpha(c\,\sigma))$$

Replacing α with $\alpha \circ c$, however, reduces us to the already proven case of $C = \Delta^n, D = \mathcal{S}$. □

Lemma 59 is already powerful. However, it does not capture that this equivalence is *natural* in both c and X—or, more precisely, since c is $\flat$-annotated and the equivalence is in $\mathcal{U}$, the naturality it yields is trivial. We are able to prove a far stronger version of the Yoneda lemma that (1) does not need to assume that $c :_\flat C$, and (2) yields the desired functoriality in both c and X. To do so, we replace $\mathsf{hom}(-,-)$ with Φ:

Theorem 62 (Functorial Yoneda lemma)**.** *There is a natural isomorphism* $\Phi_{\widehat{C}}(\mathsf{Yo}^\dagger(-),-) \cong \mathsf{eval} : \mathsf{C_{op}} \times \widehat{C} \to \mathcal{S}$.

***Remark* 63.** This result uses a handful of results from Section 4. These forward references are justified: we do not use Theorem 62 till Section 4.3. We present the proof here for conceptual coherence.

Доведення. The central difficulty in this proof is to find a map $\Phi_{\widehat{C}}(\mathsf{Yo}^\dagger(-),-) \to \mathsf{eval}$ which can then be checked to be an equivalence. To construct this map, we use the presentation of $\mathsf{C_{op}} \times \widehat{C} \to \mathcal{S}$ as covariant families over $\mathsf{C_{op}} \times \widehat{C}$. In particular, we consider the following pullback diagrams:

$$\begin{tikzcd}
\tilde{\Phi}_C \arrow[r, "v"] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & V \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \widehat{\mathsf{C}}_{\mathsf{tw}} \arrow[d] \\
\mathsf{C_{op}} \times C \arrow[r, "\mathsf{id} \times \mathsf{Yo}"'] & \mathsf{C_{op}} \times \widehat{C} \arrow[r, "\mathsf{Yo}^\dagger \times \mathsf{id}"'] & \widehat{\mathsf{C}}_{\mathsf{op}} \times \widehat{C}
\end{tikzcd}$$

$$
\begin{tikzcd}
\mathsf{C}_{\mathsf{tw}} \arrow[r, "w"] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & W \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \sum_{A:\mathcal{S}} A \arrow[d] \\
\mathsf{C}_{\mathsf{op}} \times C \arrow[r, "\mathsf{id} \times \mathsf{Yo}"'] & \mathsf{C}_{\mathsf{op}} \times \widehat{C} \arrow[r, "\mathsf{eval}"'] & \mathcal{S}
\end{tikzcd}
$$

The claim is then that $V \mathsf{Equiv} W$. To show this, we argue that if we replace the composite $\mathsf{C}_{\mathsf{tw}} \to \mathsf{C}_{\mathsf{op}} \times C \to \mathsf{C}_{\mathsf{op}} \times \widehat{C}$ with the free covariant family, then the maps $\bar{v}, \bar{w}$ induced by v and w are both equivalences. The conclusion then shows $\bar{v} \circ \bar{w}^{-1}$ is the desired equivalence.

Using [47, Theorem 2.41] there is a free covariant fibration $Z : \mathsf{C}_{\mathsf{op}} \times \widehat{C} \to \mathcal{U}$ and if $c :_\flat C$ and $X :_\flat \widehat{C}$ then by Corollary 73 we have the following:

$$
\begin{aligned}
&Z(c, X) = \\
&\quad \bigcirc_{\mathsf{grpd}} \big(\textstyle\sum_{c_0, c_1} \mathsf{hom}(c_0, c) \times \mathsf{hom}(\mathsf{Yo} c_1, X) \times \Phi(c_0, c_1)\big)
\end{aligned}
$$

To show that e.g., v induces an equivalence, we must show that the following map is an equivalence:

$$
Z(c, X) \to \mathsf{hom}(\mathsf{Yo}^\dagger(c), X)
$$

We may use Theorem 61 and assume that there exists $c' :_\flat C$ such that $c = \mathsf{mod}_{\mathsf{op}}(c')$ and that $X :_\flat \widehat{C}$. Moreover, since the right-hand side is a groupoid, this map is uniquely induced by extending the canonical map of the following type:

$$
\begin{aligned}
&\big(\textstyle\sum_{c_0, c_1} \mathsf{hom}(c_0, c) \times \mathsf{hom}(\mathsf{Yo} c_1, X) \times \Phi(c_0, c_1)\big) \\
&\quad \to \Phi(\mathsf{Yo}^\dagger(c), X) \mathsf{Equiv} X(c')
\end{aligned}
$$

This map sends (c_0, c_1, f, α, t) to $\alpha\, c\, (\Phi(f, \mathsf{id})\, t)$ and one may check directly that the assignment $x \mapsto \eta(c, c', \mathsf{id}, \mathsf{id}, F_x)$ is a quasi-inverse to this map where $F_x : \mathsf{hom}(\mathsf{Yo}(c'), X)$ corresponds to $x : X(\mathsf{mod}_{\mathsf{op}}(c'))$ under Lemma 59. The case for w is similar. □

4 Revisiting adjunctions

With presheaves and the Yoneda embedding available, we now revisit the theory of adjoint functors introduced by [46] in STT. They define a pair of functions $f : C \to D$ and $g : D \to C$ to be adjoint when equipped with $\iota : \prod_{c,d} \mathsf{hom}(f(c), d)\mathsf{Equiv}\mathsf{hom}(c, g(d))$. While they produce several equivalent reformulations using a unit and counit natural transformations, no non-trivial examples of adjunctions are given—unsurprisingly, since concrete examples of categories in STT are relatively recent. Even with $\mathcal{S}$ available it is quite difficult to produce examples of such adjunctions.

It is far more feasible to construct only f and then show that $\Phi(f^\dagger(-), d) : \widehat{C}$ is representable for every $d :_\flat D$. This is comparable to Theorem 61: we wish to give a functorial definition of either f or g and a *non-functorial* definition of the other, and then show that this can be upgraded to a full adjunction. In this section, we show that this is indeed possible, and we observe that a number of important adjunctions and results are then immediately within reach. In particular, we shall use this technique to prove that $\widehat{C}$ is cocomplete and, moreover, is the free cocompletion of C.

4.1 Pointwise adjunctions to adjunctions

Let us begin by formalizing the notion of pointwise adjoints:

Definition 64. We say that $f :_\flat C \to D$ is a pointwise left adjoint if the following type is inhabited:

$$\prod_{d :_\flat D} \mathsf{isRepr}(\Phi(f^\dagger(-), d))$$

Dually, f is a pointwise right adjoint if $f^\dagger : \mathsf{C}_{\mathsf{op}} \to \mathsf{D}_{\mathsf{op}}$ is a pointwise left adjoint.

Our main theorem relies on two crucial preliminary results. The first shows that any pointwise left adjoint f gives rise to a function in the other direction picking out the various (necessarily unique) representing objects for $\Phi(f^\dagger(-), d)$.

Lemma 65. *If $f :_\flat C \to D$ is a pointwise left adjoint, then the type of morphisms $g :_\flat D \to C$ equipped with a natural isomorphism $\iota : \Phi(f^\dagger(-), -) \cong \mathsf{Yo} \circ g$ is contractible.*

Доведення. Since Yo is an embedding, this type is a proposition. It therefore suffices to show that it is inhabited. By assumption, $\bar{g} = \Phi(f^\dagger(-), d)$ is representable for all $d :_\flat D$, and thus it factors through $\widehat{C}_{\mathsf{isRepr}}$. Post-composing with the equivalence $\widehat{C}_{\mathsf{isRepr}}\mathsf{Equiv}C$ yields the desired $g : D \to C$. □

Using this, we prove a universal case of the theorem improving a pointwise adjoint to an adjoint: every $g :_\flat D \to C$ that is a cartesian fibration [10] such that the fiber over every $c :_\flat C$ has an initial object [4] admits a left adjoint.

Lemma 66. *If $g :_\flat D \to C$ is cartesian and for each $c :_\flat C$ the fiber D_c has an initial object, then there exists $f : C \to D$ such that $f(c)$ is initial in D_c for all $c : C$.*

Доведення. Note that $\mathsf{hasInitialObj}(D_c)$ is a proposition and, therefore, by Axiom Д we may assume $\mathsf{hasInitialObj}(D_c)_\flat$ holds for each $c :_\flat C$. With this observation to hand, we can show that g is a pointwise right adjoint: if $c :_\flat C$, $d : D$:

$$\begin{aligned} \Phi_C(\mathsf{mod_{op}}(c), g(d)) & \mathsf{Equiv} \mathsf{hom}(c, g(d)) \\ & \mathsf{Equiv} \mathsf{hom}(0_{D_c}, d) && g \text{ is cartesian} \\ & \mathsf{Equiv} \Phi_D(\mathsf{mod_{op}}(0_{D_c}), d) \end{aligned}$$

In this last step, we use our observation that $\mathsf{hasInitialObj}(D_c)_\flat$ and, crucially, that not only $\mathsf{hasInitialObj}(D_c)$ holds. In particular, we rely on the fact that $0_{D_c} :_\flat D_c$.

Accordingly, we obtain a function $f :_\flat C \to D$ which sends $c :_\flat C$ to 0_{D_c}. It remains to show that $f(c)$ is initial in D_c for all $c : C$. Since $D = \sum_{c:C} D_c$, this amounts to the following map being an equivalence: $\left(\sum_{d:D} \mathsf{hom}_{D_{g(d)}}(f(g(d)), d)\right) \to D$.

To prove this, we use Theorem 44 which allows us to reduce to the $\flat$-annotated case, where the conclusion follows from the fact that $f(c)$ is then initial in D_c. □

Theorem 67. *Pointwise right adjoints are right adjoints.*

Доведення. Given a map $g :_\flat D \to C$, consider the cartesian family

$$\pi : C \downarrow g = \left(\textstyle\sum_{c:C} \sum_{d:D} \mathsf{hom}(c, g(d))\right) \to C$$

Since g is a pointwise right adjoint, each fiber of π over $c :_\flat C$ has an initial object. We then apply Lemma 66 to obtain $\bar{f} : C \to C \downarrow g$. Finally, the composite $\pi_2 \circ \bar{f}$ is the desired left adjoint to g:

$$\begin{aligned} & \mathsf{hom}_C(c, g(d)) \\ & \quad \mathsf{Equiv} \textstyle\sum_{\alpha:\mathsf{hom}_C(c,g(d))} \mathsf{hom}_{C\downarrow g_c}(\bar{f}(c), (c, d, \alpha)) \end{aligned}$$

$$\mathsf{Equiv}\sum_{\alpha:\mathsf{hom}_C(c,g(d))}\sum_{\beta:\mathsf{hom}_D(f(c),d)} g(\beta)\circ\pi_3(\bar{f}(c)) = \alpha$$
$$\mathsf{Equiv}\mathsf{hom}_D(f(c),d)$$

The first step uses the initiality of $\bar{f}(c)$ in the fiber over c and the second unfolds the definition of a morphism in $C\downarrow g$. □

4.2 Examples of adjunctions

We take advantage of Theorem 67 to produce vital examples of adjoints.The most important is the following:

Theorem 68. *If $f :_\flat D \to C$ and D is small, then $\widehat{f} :- (f^\dagger)^* : \widehat{C} \to \widehat{D}$ is a right adjoint with left adjoint $f_!$.*

Доведення. For notational simplicity, we replace C and D with C_{op} and D_{op}. By Theorem 67, it suffices to assume $X :_\flat D \to \mathcal{S}$ and to construct $f_!(X) : C \to \mathcal{S}$ along with a natural bijection $\prod_Y \mathsf{hom}(f_!(X), Y)\mathsf{Equiv}\mathsf{hom}(X, f^*(Y))$. This is an immediate consequence of [47, Theorem 2.41] after localizing the composite $\sum_{c:C} X(C) \to D$ against the map $\{0\} \to \mathbb{I}$. □

Corollary 69. *The left adjoint $f_! \dashv \widehat{f}$ satisfies $f_! \circ \mathsf{Yo} \cong \mathsf{Yo} \circ f$.*

Corollary 70. *$\mathcal{S}$ is small cocomplete: $\mathsf{const} : \mathcal{S} \to \mathcal{S}^C$ is a right adjoint with left adjoint* colim_ *for small categories $C :_\flat \mathcal{U}$. Explicitly, if $X :_\flat C \to \mathcal{S}$, then* colim_ $CX = \bigcirc_{\mathsf{grpd}}\left(\sum_{c:C} X(c)\right)$.

***Remark* 71.** One can prove $\mathcal{S}$ is complete (that $\mathsf{const} \dashv \lim\_$) by a result of [19]. In particular, they show $\prod_{c:C} X(c) : \mathcal{S}$ whenever $X : C \to \mathcal{S}$ and $C :_\flat \mathcal{U}$. Corollary 48 and Lemma 51 then imply that $\mathsf{hom}\left(A, \prod_{c:C} X(c)\right)\mathsf{Equiv}\mathsf{hom}(\mathsf{const}\, A, X)$.

Lemma 72. *If $c :_\flat C$ then $\widehat{c} : \widehat{C} \to \mathcal{S}$ is a left adjoint.*

Доведення. For simplicity, we once more replace C with C_{op} and invoke Theorem 67. We then assume that we are given $X :_\flat \mathcal{S}$ and define $c_* X$ to be $\lambda c'.\, X^{\mathsf{hom}(c',c)}$. This is covariant in c' [19]. It then suffices to show that the map $X^{\mathsf{hom}(c,c)} \to X$ given by evaluation at the identity map induces an equivalence $\mathsf{hom}_{C\to\mathcal{S}}(Y, c_* X)\mathsf{Equiv}\mathsf{hom}_{\mathcal{S}}(Y(c), X)$. This, in turn, is a consequence of the fact that the map $Y(c) \to \bigcirc_{\mathsf{grpd}}\left(\sum_{c':C} \mathsf{hom}(c', c) \times Y(c')\right)$ is an equivalence, so we may decompose the evaluation map as follows:

$$\mathsf{hom}_{C\to\mathcal{S}}(Y, c_* X)$$

$$\begin{aligned}&\mathsf{Equiv}\textstyle\prod_{c':C} Y(c') \to X^{\mathsf{hom}(c',c)}\\&\mathsf{Equiv}\textstyle\prod_{c':C}\mathsf{hom}(c',c)\times Y(c')\to X\\&\mathsf{Equiv}\bigcirc_{\mathsf{grpd}}\left(\textstyle\sum_{c':C}\mathsf{hom}(c',c)\times Y(c')\right)\to X\\&\mathsf{Equiv}Y(c)\to X\end{aligned}$$

□

Combining Theorem 68 and Lemma 72, we obtain the following characterization of $f_!X$:

Corollary 73. *If $X :_\flat \widehat{C}$, $f :_\flat C \to D$, and $d :_\flat D$ then we may explicitly identify $(f_!X)\,d$ as the following type:*

$$\bigcirc_{\mathsf{grpd}}\left(\textstyle\sum_{c:\mathsf{C}_{\mathsf{op}}} X(c)\times\mathsf{hom}(f^\dagger c, d)\right)$$

Доведення. We first observe that $\widehat{d}f_!(X) = (f_!X)\,d$. Transposing, we have $\mathsf{hom}(\widehat{d}f_!X, Z)\mathsf{Equiv}\mathsf{hom}(X, \widehat{f}d_*Z)$ for every $Z :_\flat \mathcal{S}$. We calculate $\mathsf{hom}(X, \widehat{f}d_*Z)$ using the definition of d_* provided above:

$$\begin{aligned}&\mathsf{hom}_{\widehat{C}}(X, f^*d_*Z)\\&\mathsf{Equiv}\textstyle\prod_{c:\mathsf{C}_{\mathsf{op}}} X(c)\to\mathsf{hom}(f^\dagger c, d)\to Z\\&\mathsf{Equiv}\left(\textstyle\sum_{c:\mathsf{C}_{\mathsf{op}}} X(c)\times\mathsf{hom}(f^\dagger c, d)\right)\to Z\end{aligned}$$

Therefore $\widehat{d}f_!X$ satisfies the universal property of $\bigcirc_{\mathsf{grpd}}\left(\sum_{c:\mathsf{C}_{\mathsf{op}}} X(c)\times\mathsf{hom}(f^\dagger c, d)\right)$

□

The following lemma does not require Theorem 67, but is merely a consequence of manipulating natural transformations:

Lemma 74. *If $f : C \to D$ is an adjoint so is $f_* : C^A \to D^A$.*

Corollary 75. *If C is (co)complete so is C^D and (co)limits are computed pointwise. In particular, $\widehat{C}$ is (co)complete.*

Corollary 76. *The Yoneda embedding preserves all limits.*

Доведення. If $F :_\flat I \to C$ and $\lim\_F$ exists, then functoriality of Yo induces a map $\mathsf{Yo}\lim\_F \to \lim\_(\mathsf{Yo}\circ F)$, so it suffices to check that this map is invertible at all $c :_\flat C$. Unfolding, we must argue that $\mathsf{hom}(c, \lim\_F)\mathsf{Equiv}\lim\_\mathsf{hom}(c, F)$ is an equivalence, but this is immediate by Lemma 51. □

Finally, we show the full subcategories $\mathcal{S}_{\leq n}$ of $\mathcal{S}$ defined by n-truncated types form reflective subcategories of $\mathcal{S}$. The idea is simple: use the truncation HITs. However, it is not automatic that they restrict to $\|-\|_n : \mathcal{S} \to \mathcal{S}_{\leq n}$. We prove this alongside with the reflectivity of $\mathcal{S}_{\leq n}$ using Theorem 67:

Corollary 77. *The inclusion $\mathcal{S}_{\leq n} \to \mathcal{S}$ is a right adjoint.*

Corollary 78. *$\mathcal{S}_{\leq n}$ is (co)complete.*

The same methodology applies to the subcategory of modal types associated to an idempotent monad [47].

Example 79 (Isbell conjugation)**.** If $C :_\flat \mathcal{U}$, then the Isbell conjugation map ϕ is a left adjoint:

$$\begin{aligned}
&\phi : \widehat{C} \to \mathsf{C} \to \mathcal{S}_{\mathsf{op}} \\
&\phi(X) = \mathsf{mod}_{\mathsf{op}}(\lambda c.\, \Phi(X, \mathsf{Yo} c))
\end{aligned}$$

4.3 The universal property of presheaf categories

Next, we generalize Theorem 68 to show that if $f :_\flat C \to E$ where C is a small category and E is a cocomplete category, then $\Phi(f^\dagger(-), -) : E \to \widehat{C}$ is a right adjoint loosely following the argument given by [13]. We begin with a few general lemmas. In what follows, fix C and E as above.

First, as a corollary of the proof of Theorem 68:

Lemma 80. *The colimit of $\mathsf{Yo} : C \to \widehat{C}$ is $1\widehat{C} = \lambda\_.\, 1$.*

From the above, and further inspection of colimits, we are able to derive a result of independent interest: Every presheaf is the colimit of representable presheaves.

Lemma 81 (Density of Yo)**.** *If $X :_\flat \widehat{C}$, then $X \cong \mathrm{colim}\_\widetilde{\mathsf{X}}_{\mathsf{op}} \mathsf{Yo} \circ \pi^\dagger$, where $\widetilde{X} = \sum_{c : \mathsf{C}_{\mathsf{op}}} X(c)$.*

Доведення. We begin with the following computation where $\pi : \widetilde{X} \to \mathsf{C}_{\mathsf{op}}$ and $\pi_!^\dagger : \mathcal{S}^{\widetilde{X}} \to \widehat{C}$:

$$\pi_!^\dagger 1 \cong \pi_!^\dagger(\mathrm{colim}\_\widetilde{\mathsf{X}}_{\mathsf{op}} \mathsf{Yo}) \cong \mathrm{colim}\_\widetilde{\mathsf{X}}_{\mathsf{op}} \pi_!^\dagger \circ \mathsf{Yo} \cong \mathrm{colim}\_\widetilde{\mathsf{X}}_{\mathsf{op}} \mathsf{Yo} \circ \pi^\dagger$$

We have used the fact that $\pi_!^\dagger$, a left adjoint, commutes with colimits [4]. To show $\pi_!^\dagger 1 \cong X$, we note that for all $Z : \widehat{C}$:

$$\mathsf{hom}(\pi_!^\dagger 1, Z)\mathsf{Equiv}\mathsf{hom}_{\widetilde{X} \to \mathcal{S}}(1, Z \circ \pi)$$

$$\begin{aligned}&\mathsf{Equiv}\prod_{(c,x):\sum_{c:\mathsf{C_{op}}}X(c)}Z(c)\\&\mathsf{Equiv}\prod_{c:\mathsf{C_{op}}}X(c)\to Z(c)\\&\mathsf{Equiv}\mathsf{hom}(X,Z)\end{aligned}$$

The conclusion now follows from the Yoneda lemma. □

Lemma 82. $\mathsf{N}_f = \Phi(f^\dagger(-), -) : E \to \widehat{C}$ *is a right adjoint.*

Доведення. We will prove that N_f is a pointwise right adjoint. Accordingly, fixing $X :_\flat \widehat{C}$ we must construct $e : \mathsf{E_{op}}$ such that $\Phi(e, -) \cong \Phi(\mathsf{mod_{op}}(X), \mathsf{N}_f(-))$. Since $X \cong$ colim_$\widetilde{\mathsf{X}}_{\mathsf{op}}\mathsf{Yo} \circ \pi^\dagger$ and E is cocomplete, by the dual of Corollary 76 it suffices to assume $\mathsf{mod_{op}}(X) = \mathsf{Yo}^\dagger(c)$ with $c : \mathsf{C_{op}}$.[9] Finally, take $e = f^\dagger(c)$ and $\Phi(f^\dagger(c), -) \cong \Phi(\mathsf{Yo}^\dagger(c), \mathsf{N}_f(-))$ by Theorem 62. □

We are now able to prove, as promised, the universal property of $\widehat{C}$. If we write $\mathsf{CC}(\widehat{C}, E)$ for the full subcategory of $\widehat{C} \to E$ spanned by functors preserving all colimits, then $\mathsf{Yo}^* : \mathsf{CC}(\widehat{C}, E) \to (C \to E)$ is an equivalence. To prove this, we essentially argue that there is a map sending f to the left adjoint to N_f and that this is the inverse to Yo^*.

Theorem 83. $\mathsf{Yo}^* : \mathsf{CC}(\widehat{C}, E) \to (C \to E)$ *is an equivalence.*

Доведення. We use Theorem 44. If $f :_\flat C \to E$, then $f_! : \widehat{C} \to E$ satisfies $f_! \circ \mathsf{Yo} = f$ and so Yo^* is essentially surjective:

$$\Phi((f_! \circ \mathsf{Yo})^\dagger(-), -) \cong \Phi(\mathsf{Yo}^\dagger(-), \mathsf{N}_f(-)) \cong \mathsf{N}_f = \Phi(f^\dagger(-), -)$$

Moreover, if $F :_\flat \mathsf{CC}(\widehat{C}, E)$, then $(F \circ \mathsf{Yo})_! \cong F$, so that Yo^* is a bijection on $\flat$-elements. Let us write $f = F \circ \mathsf{Yo}$. We first construct a comparison map $\mathsf{hom}(f_!, F)$ by constructing a natural transformation $\mathsf{hom}(\mathsf{id}, \mathsf{N}_f(F(-)))$. Currying, this is equivalent to constructing a natural transformation between maps $\mathsf{C_{op}} \times \widehat{C} \to \mathcal{S}$ and, in this form, id is given by evaluation ϵ and $\mathsf{N}_F(F(-))$ is $\Phi(f(-), F(-))$. We can replace ϵ with $\Phi(\mathsf{Yo}(-), -)$ by Theorem 62 and $\Phi(f(-), F(-)) = \Phi(F(\mathsf{Yo}(-)), F(-))$ by definition. Accordingly, the relevant map is supplied by Φ_F. It is routine to check that this is pointwise an equivalence by Lemma 81.

[9] Note the lack of $\flat$-annotation here: we must ensure that we are functorial in c in order to obtain a *diagram* in E.

Finally, we now show that Yo^* is fully faithful. To show that it is fully faithful, we must show that if $f, g :_\flat C \to E$, then $\mathsf{hom}(f_!, g_!) \cong \mathsf{hom}(f, g)$. Both sides are groupoids, so it suffices to consider $\flat$-annotated elements. If $\alpha :_\flat \mathsf{hom}(f, g)$, then by transposing we may regard α as an element of $\mathsf{C} \to \mathsf{E}^{\mathbb{I}}{}_\flat$ and the previous observation ensures that this type is equivalent to $\mathsf{CC}(\widehat{C}, E^{\mathbb{I}})_\flat$ which yields the desired conclusion after transposing. □

5 The theory of Kan extensions

A unifying concept in category theory are *Kan extensions*, which are universal extensions of functors along functors on the same domain. Mac Lane, one of the founders of category theory, famously stated: "The notion of Kan extensions subsumes all the other fundamental concepts of category theory," such as (co)limits and adjunctions [28, 43].

Definition 84 (Kan extensions)**.** Given a map $f : C \to D$ and a category E, the left (right) Kan extension $\mathsf{Lan} f$ ($\mathsf{Ran} f$) is the left (right) adjoint to $f^* : E^D \to E^C$.

While the definition makes sense in general, to use the results of the previous sections, we shall assume $f :_\flat C \to D$ and $E :_\flat \mathcal{U}$. In Section 5.1 we show that Kan extensions exist whenever E is (co)complete and in Sections 5.2 and 5.3 we put this to work by deducing two important results: Quillen's theorem A and the properness of cocartesian fibrations. Our arguments for the existence of Kan extensions and Quillen's theorem A adapt the (model-agnostic) ∞-categorical arguments of [42].

5.1 Existence and characterization of Kan extensions

We can prove that Kan extensions can be computed in an expected way. For $d : D$, we write $C_{/d} :\equiv C \times_D D_{/d}$ and $C_{d/} :\equiv C \times_D D_{d/}$. We assume that C and D are both small so each $C_{/d}$ is also small. By Theorem 68 and Lemma 74:

Lemma 85. *If $E = \widehat{A}$ for some category $A :_\flat \mathcal{U}$, then $\mathsf{Lan} f$ exists. Moreover, if $X :_\flat C \to E$ and $d :_\flat D$, then $\mathsf{Lan} f\, X\, d = \mathrm{colim\_}(C_{/d} \to C \to E) = \bigcirc_{\mathsf{grpd}}\left(\sum_{(c,\_):C_{/d}} X(c)\right)$.*

This yields more generally:

Theorem 86. *If E is cocomplete, then* $\mathsf{Lan}f$ *exists, and if* $X :_\flat C \to E$*,* $d :_\flat D$*, then* $\mathsf{Lan}f\, X\, d = \mathrm{colim\_}(C_{/d} \to C \to E)$*.*

Доведення. It suffices to argue that precomposition is pointwise a right adjoint and so we fix $X :_\flat C \to E$. By Theorem 83, we may view X as the composition $\bar{X} \circ \mathsf{Yo}$, where $\bar{X} : \widehat{C} \to E$ is the left adjoint to N_X. Next, we observe by Lemma 85 that $\mathsf{Yo} : C \to \widehat{C}$ admits an extension to D along f, namely $\mathsf{Lan}f\mathsf{Yo} : D \to \widehat{C}$, and we claim that $\bar{X} \circ \mathsf{Lan}f\mathsf{Yo}$ is our desired extension of f. Fixing $Z : D \to E$, we calculate:

$$\begin{aligned}
\mathsf{hom}_{D\to E}(\bar{X} \circ \mathsf{Lan}f\mathsf{Yo}, Z) & \mathsf{Equiv}\mathsf{hom}_{D\to\widehat{C}}(\mathsf{Lan}f\mathsf{Yo}, \mathsf{N}_X \circ Z) \\
& \mathsf{Equiv}\mathsf{hom}_{C\to\widehat{C}}(\mathsf{Yo}, \mathsf{N}_X \circ Z \circ f) \\
& \mathsf{Equiv}\mathsf{hom}_{C\to E}(\bar{X} \circ \mathsf{Yo}, Z \circ f) \\
& = \mathsf{hom}_{C\to E}(X, Z \circ f)
\end{aligned}$$

The expected colimit formula continues to hold as a consequence of Lemma 85 and the cocontinuity of $\bar{X}$. □

By duality, we obtain the following variant:

Theorem 87. *If E is complete, then* $\mathsf{Ran}f$ *exists and is specified by the dual limit formula:* $\mathsf{Ran}f\, X\, d = \mathrm{lim\_}(C_{d/} \to C \to E)$*.*

5.2 Final and initial functors

It is frequently useful to show that the limit of a complex diagram D can be calculated by first restricting to a simpler diagram using $f : C \to D$ and calculating the limit there e.g., restricting from $\mathbb{Z}$ to $\mathbb{Z}_{\leq 0}$. When this approach is valid, f is said to be initial:

Definition 88. A functor $f :_\flat C \to D$ is *initial* if for every $X :_\flat D \to \mathcal{S}$ the map $\mathrm{lim\_}DX \to \mathrm{lim\_}CX \circ f$ is an equivalence. A map is *final* if its opposite is initial.

While this definition is asymmetrical in its treatment of initiality and finality, we shall restore the symmetry as a consequence of Quillen's Theorem A in the next section, see Corollary 99.

Recall that $\mathrm{lim\_}DX = \prod_{d:D} X(d)$ and so the definition of initiality equivalently states that the restriction map $\left(\prod_{d:D} X(d)\right) \to \left(\prod_{c:C} X(f(c))\right)$ is an equivalence.

Example 89. The $\{0\}/\{1\}$ inclusion $\mathbf{1} \to \mathbb{I}$ is initial/final.

Lemma 90. *If $f :_\flat C \to D$ is initial and $X :_\flat D \to E$, then* $\lim\_C(X \circ f)$ *and* $\lim\_DX$ *both exist whenever either exists and are canonically isomorphic.*

Доведення. By Corollary 76, we replace E with $\widehat{E}$ and by Corollary 75 we reduce to $\mathcal{S}$ where the result is immediate. $\square$

Lemma 91. *If $C :_\flat \mathcal{U}$, then $\bigcirc_{\mathsf{grpd}} C\mathsf{Equiv} \bigcirc_{\mathsf{grpd}} \mathsf{C}_{\mathsf{op}}$.*

Доведення. We observe that $\bigcirc_{\mathsf{grpd}} C\mathsf{Equiv}\bigcirc_{\mathsf{grpd}} \mathsf{C}_\flat$ and likewise for C_{op}. Accordingly, we note that:

$$\begin{gathered}\mathsf{C}_{\mathsf{op}} \to \mathsf{X}_{\flat_\flat}\mathsf{Equiv}\bigcirc_{\mathsf{grpd}} \mathsf{C}_{\mathsf{op}} \to \mathsf{X}_{\flat_\flat} \\ \mathsf{C} \to \mathsf{X}_{\flat\flat}\mathsf{Equiv}\bigcirc_{\mathsf{grpd}} \mathsf{C} \to \mathsf{X}_{\flat_\flat} \\ \mathsf{C} \to \mathsf{X}_{\flat\flat}\mathsf{Equiv}\mathsf{C}_{\mathsf{op}} \to \mathsf{X}_{\flat_\flat}\end{gathered}$$

Finally, the result follows from a simple Yoneda argument. $\square$

Lemma 92. *For every C, the canonical map $C \to \bigcirc_{\mathsf{grpd}} C$ is both initial and final.*

Доведення. By Lemma 91, it suffices to argue that this map is initial. To this end, we must show the following map to be an equivalence for every $X : \bigcirc_{\mathsf{grpd}} C \to \mathcal{S}$:

$$\left(\prod_{d:\bigcirc_{\mathsf{grpd}} C} X(d)\right) \to \left(\prod_{c:C} X(\eta(c))\right)$$

However, $X(d)$ is discrete for every $d : \bigcirc_{\mathsf{grpd}} C$ and so this is simply the universal property of $\bigcirc_{\mathsf{grpd}}$. $\square$

Corollary 93. *If $\bigcirc_{\mathsf{grpd}} C = \mathbf{1}$, then* $\lim\_CA = A$ *for $A : \mathcal{S}$.*

5.3 Quillen's Theorem A

Our next goal is to prove the ∞-categorical version of Quillen's theorem A. Unlike traditional proofs, we follow [42] and rely on having already established the basic apparatus of Kan extensions to simplify our argument.

Definition 94. A functor $f :_\flat C \to D$ is Quillen final if $\bigcirc_{\mathsf{grpd}}(C_{d/})\mathsf{Equiv}\mathbf{1}$ for all $d :_\flat D$

Theorem 95. *A functor $f :_\flat C \to D$ is final if and only if it is Quillen final.*

***Remark* 96.** This result shows that, in particular, finality doesn't depend on the particular universe $\mathcal{S}$ chosen.

Lemma 97. *If $f :_\flat C \to D$ is Quillen final and $X :_\flat D \to \widehat{A}$, then* colim_ DXEquivcolim_ $CX \circ f$*.*

Доведення. This statement is pointwise, so we quickly reduce to $\mathcal{S}$ instead of $\widehat{A}$. In this situation, we wish to show that the following commutes:

$$\begin{tikzcd} \mathcal{S}^D \arrow[rr, "f^*"] \arrow[dr, "\mathrm{colim}_D"'] & & \mathcal{S}^C \arrow[dl, "\mathrm{colim}_C"] \\ & \mathcal{S} & \end{tikzcd}$$

Note that all three morphisms are left adjoints, and so it suffices to compare their right adjoints: the constant functors Δ_C and Δ_D, along with the right Kan extension $\mathsf{Ran} f$. We next note that there is at least a comparison map $\Delta_D \to \mathsf{Ran} f \circ \Delta_C$ given by transposing the identity map $f^* \circ \Delta_D \to \Delta_C$. We must argue that this map is pointwise invertible, and so we reduce to considering $X :_\flat \mathcal{S}$ and $d :_\flat D$, and we must show the following, using Theorem 87: XEquivlim_ $C_{d/}X$. This now follows from our assumption and Lemma 93. □

Lemma 98. *If $f :_\flat C \to D$ is Quillen final, E a cocomplete category, and $X :_\flat D \to E$, then* colim_ DXEquivcolim_ $CX \circ f$*.*

Доведення. We reduce to the case where $E = \widehat{D}$ (and therefore Lemma 97) by factoring X as $\bar{X} \circ \mathsf{Yo}$ and noting that $\bar{X}$ preserves colimits by construction. □

Proof of Theorem 95. To see that Quillen finality implies finality, we apply Lemma 98 to $\mathcal{S}_{\mathsf{op}}$, and calculate:

$$\text{lim\_ } \mathsf{D}_{\mathsf{op}} X \mathsf{Equiv} \text{colim\_ } DX^\dagger \mathsf{Equiv} \text{colim\_ } CX^\dagger \circ f \mathsf{Equiv} \text{lim\_ } \mathsf{C}_{\mathsf{op}} X \circ f^\dagger$$

For the reverse, suppose that f is final. We note that by the dual of Lemma 90 (again applied to $\mathcal{S}_{\mathsf{op}}$), the canonical map colim_ $CX \circ f \to$ colim_ DX is an equivalence for any $X :_\flat D \to \mathcal{S}$. Fix $d :_\flat D$ and choose $X = \hom(d, -) = \Phi(\mathsf{mod}_{\mathsf{op}}(d), -)$ such that the colimits in question are precisely $\bigcirc_{\mathsf{grpd}} D_{d/}$ and $\bigcirc_{\mathsf{grpd}} C_{d/}$, using Theorem 68. This completes the proof since $\bigcirc_{\mathsf{grpd}} D_{d/} = \mathsf{1}$. □

Corollary 99. *A functor* $f :_\flat C \to D$ *is final if and only if, for every* $X :_\flat D \to \mathcal{S}$ *the map* $\mathrm{colim\_}DX \to \mathrm{colim\_}CX \circ f$ *is an equivalence.*

This restores the symmetry between initial and final functors, as promised. We offer another symmetric definition of initiality and finality, informed by [12, Ch. 8].

Definition 100 (Covariant equivalences)**.** Fix $p :_\flat C \to A$ and $q :_\flat D \to A$ between categories A, C, D. Let $f :_\flat C \to D$ be a fibered map as follows:

$$\begin{tikzcd} C \arrow[rr, "f"] \arrow[dr, "p"'] & & D \arrow[dl, "q"] \\ & A & \end{tikzcd}$$

We call f a *covariant equivalence* if for all families $X :_\flat A \to \mathcal{S}$ reindexing gives rise to an equivalence, i.e.,:

$$f^* : \left(\prod_{a:A} D_a \to X_a\right) \to \left(\prod_{a:A} C_a \to X_a\right)$$

Dually, f is called *contravariant equivalence* precomposition with respect to all contravariant families is an equivalence.

Lemma 101. *Let* f *as below be a covariant equivalence with respect to* p *and* q*. Then, for any functor* $r :_\flat B \to A$ *it is also a covariant equivalence with respect to* rp *and* rq*:*

$$\begin{tikzcd} C \arrow[rr, "f"] \arrow[dr, "p"'] & & D \arrow[dl, "q"] \\ & A \arrow[d, "r"] & \\ & B & \end{tikzcd}$$

Доведення. We get the following induced square:

$$\begin{tikzcd} \left(\prod_{b:B} D_b \to X_b\right) \arrow[r, "f^*", "\simeq"'] \arrow[d, "\simeq"'] & \left(\prod_{b:B} C_b \to X_b\right) \arrow[d, "\simeq"] \\ \left(\prod_{a:A} D_a \to X_{r(a)}\right) \arrow[r, "f^*"'] & \left(\prod_{a:A} C_a \to X_{r(a)}\right) \end{tikzcd}$$

The upper horizontal map is an equivalence by the preconditions. The goal is to show that the lower horizontal map is an equivalence, too. But this follows from 3-for-2 for equivalences. □

Lemma 102 (Characterizations of initiality)**.** *Let $f :_\flat C \to D$ be a functor. Then the following are equivalent:*

(1) *f is initial.*

(2) *Let $X :_\flat A \to \mathcal{S}$ be a family with associated left fibration $\pi :_\flat \widetilde{X} \to A$. Then any square of the following form has a filler $\overline{\varphi}$, uniquely up to homotopy:*

$$\begin{tikzcd} C \arrow[r, "\varphi"] \arrow[d, "f"'] & \widetilde{X} \arrow[d, "\pi"] \\ D \arrow[r, "\alpha"'] \arrow[ur, dashed, "\overline{\varphi}"] & A \end{tikzcd}$$

(3) *For any family $X :_\flat A \to \mathcal{S}$ the following square is a pullback:*

$$\begin{tikzcd} \widetilde{X}^D \arrow[r, "l_jon_tikz_diagram_north"] \arrow[d, "l_jon_tikz_diagram_west"'] \arrow[dr, phantom, very near start, "\lrcorner"] & \widetilde{X}^C \arrow[d, "l_jon_tikz_diagram_east"] \\ A^D \arrow[r, "l_jon_tikz_diagram_south"'] & A^C \end{tikzcd}$$

(4) *f is a covariant equivalence with respect to any $\alpha :_\flat D \to A$.*

The analogous characterization holds for contravariant equivalences and final functores.

Доведення. Conditions (2) and (3) are readily seen to be equivalent by commuting $\prod$ and $\sum$. Condition (4) unfolds to the following: for any $X :_\flat A \to \mathcal{S}$ reindexing along f is an equivalence, namely

$$f^* : \left(\prod_{a:A} D_a \to X_a\right) \xrightarrow{\sim} \left(\prod_{a:A}(\sum_{d:D_a} C_{a,d}) \to X_a\right)$$

This, again, is readily seen to be equivalent to (2).

We turn to the implication (4) $\implies$ (1). But this is clear, since (1) says that f is a covariant equivalence with respect to itself and $\mathsf{id}D$.

For the converse direction (1) $\implies$ (4) we use the insight just made together with Lemma 101. ◻

The following alternative characterization is also often useful:

Lemma 103. *$f :_\flat C \to D$ is initial (resp. final) if and only if for every covariant (resp. contravariant) family $\pi :_\flat X \to Y$, f is left orthogonal to π, i.e.,* $\mathsf{isEquiv}(X^D \to X^C \times_{Y^C} Y^D)$.

Доведення. Immediate by Proposition (4) for the initial case and by duality and Corollary 99 for the final case. ◻

Corollary 104. *There is an orthogonal factorization system in the sense of [47] with the left class given by the initial functors and the right class by covariant fibrations.*

As another consequence we get the dual of Theorem 95:

Corollary 105. *A functor $f :_\flat C \to D$ is initial if and only if* $\bigcirc_{\mathsf{grpd}}(C_{/d})\mathsf{Equiv1}$ *for all $d :_\flat D$* (Quillen initial).

We demonstrate the utility of Theorem 95 by giving a new and far simpler proof that cocartesian fibrations are *proper*.

Definition 106. A functor $\pi :_\flat E \to B$ between categories is *proper* if for all pullbacks (of $\flat$-functors) of the following form, v is final if u is final:

$$\begin{tikzcd}
E'' \arrow[r, "v\, l_jon_tikz_diagram_north"] \arrow[d, "l_jon_tikz_diagram_west"'] \arrow[dr, phantom, "\lrcorner", very near start] & E' \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & E \arrow[d, "l_jon_tikz_diagram_east"] \\
B'' \arrow[r, "u\, l_jon_tikz_diagram_south"'] & B' \arrow[r] & B
\end{tikzcd}$$

We call π *smooth* if $\pi^\dagger : \mathsf{E_{op}} \to \mathsf{B_{op}}$ is proper.

Lemma 107. *Smooth and proper functors are closed under composition and pullback.*[10]

Theorem 108. *Cartesian fibrations are smooth and cocartesian fibrations are proper.*

[10]The definition of properness is formulated specifically to bake in the latter.

Доведення. It suffices to treat the proper case. Fix a cocartesian fibration $\pi :_\flat E \to B$ and note that since cocartesian fibrations are stable under pullbacks, it suffices show to that v is final in the following pullback diagram if u is final:

$$\begin{tikzcd} A \times_B E \arrow[r, "l_j on_t ikz_d iagram_n orth"] \arrow[d, "l_j on_t ikz_d iagram_w est"'] \arrow[dr, phantom, "\lrcorner", very near start] & E \arrow[d, "l_j on_t ikz_d iagram_e ast"] \\ A \arrow[r, "l_j on_t ikz_d iagram_s outh"'] & B \end{tikzcd}$$

We now use Theorem 95. For $e :_\flat E$ we compute the fiber:

$$\begin{aligned} &(A \times_B E) \times_E E_{e/} \\ &\quad \mathsf{Equiv} A \times_B E_{e/} \\ &\quad \mathsf{Equiv} A \times_B \left(\textstyle\sum_{b' : B, f : \mathsf{hom}(\pi(e), b')} (E_{b'})^{\mathbb{I}}\right) && \pi \text{ is cocartesian} \\ &\quad \mathsf{Equiv} \textstyle\sum_{(a,f) : A \times_B B_{\pi(e)/}} \left(E_{u(a)}\right)_{f_! e/} \end{aligned}$$

Applying $\bigcirc_{\mathsf{grpd}}$ to each fiber yields $\bigcirc_{\mathsf{grpd}} \left(E_{u(a)}\right)_{f_! e/} \mathsf{Equiv} \mathsf{1}$ (as coslices have initial elements) and $\bigcirc_{\mathsf{grpd}}(A \times_B B_{\pi(e)/}) \mathsf{Equiv} \mathsf{1}$ since u is final by assumption. This implies that applying $\bigcirc_{\mathsf{grpd}}$ to the entire $\sum$-type produces $\mathsf{1}$ [47]. □

Corollary 109. *If $\pi :_\flat E \to B$ is cocartesian and $X :_\flat E \to D$, then the left Kan extension $\mathsf{Lan} \pi X$ sends $b :_\flat B$ to* colim_ *$(E_b \to E \to D)$.*

5.4 Smooth and proper base change

We want to show that smooth and proper functors satisfy the *Beck–Chevalley condition*. We follow [12, Section 8.4]; see also [3] for a general discussion.

First, using the initial-covariant factorization (see 104) of a functor with small fibers we can compute the action of precomposition for (co)presheaves:

Proposition 110 ([12, Theorem 8.1.18])**.** *Let $u :_\flat A \to B$ be a functor with small fibers. Then the left adjoint $u_! \dashv u^* :_\flat \mathcal{S}^B \to \mathcal{S}^A$ acts as follows: for $F :_\flat A \to \mathcal{S}$, if*

$$\begin{tikzcd} E \arrow[r] \arrow[d, "\phi"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{S}_* \arrow[d] \\ A \arrow[r, "F"'] & \mathcal{S} \end{tikzcd}$$

denote by $E \xrightarrow{j} X \xrightarrow{\psi} B$ the initial-covariant factorization of $u \circ \phi$. Then $u_!(F)$ is the straightening of ψ:

$$\begin{tikzcd} E \arrow[r, "j"] \arrow[d, "\phi"'] & X \arrow[r] \arrow[d, "\psi"] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{S}_* \arrow[d] \\ A \arrow[r, "u"'] & B \arrow[r, "u_!(F)"'] & \mathcal{S} \end{tikzcd}$$

In particular, the unit $\eta :_\flat F \to u^* u_! F$ corresponds under directed univalence to the map $\widetilde{\eta}$:

$$\begin{tikzcd} E \arrow[r, dashed, "\widetilde{\eta}"] \arrow[rr, bend left, "j"] \arrow[dr, "\phi"'] & Y \arrow[r] \arrow[d, "u^*\psi"] \arrow[dr, phantom, "\lrcorner", very near start] & X \arrow[d, "\psi"] \\ & A \arrow[r, "u"] & B \end{tikzcd}$$

Доведення. We want to show that an equivalence is given by the map

$$\lambda\alpha.u^*\alpha \circ \eta :_\flat \hom_{B\to\mathcal{S}}(u_!F, G) \to \hom_{A\to\mathcal{S}}(F, u^*G).$$

We write the unstraightening of G as

$$\begin{tikzcd} Z \arrow[r] \arrow[d, "\gamma"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{S}_* \arrow[d] \\ B \arrow[r, "G"] & \mathcal{S} \end{tikzcd}$$

The above transposition map is equivalent to a map

$$(\prod_{b:B} \hom_{\mathcal{S}}(X_b, Z_b)) \to (\prod_{a:A} \hom_{\mathcal{S}}(E_a, (A \times_B Z)_a)) \simeq (\prod_{b:B} \hom_{\mathcal{S}}(E_b, Z_b)).$$

By directed univalence, this map corresponds to the map

$$(\prod_{b:B}(X_b \to Z_b)) \to (\prod_{b:B}(E_b \to Z_b))$$

which in turn acts as precomposition of fibered functors by j:

$$\begin{tikzcd} E \arrow[r, "j"] \arrow[dr, "u\circ\varphi"'] & X \arrow[r] \arrow[d, "\psi"'] & Z \arrow[dl, "\gamma"] \\ & B & \end{tikzcd}$$

Finally, due to Proposition 103(4) this map is an equivalence. □

Let $f :_\flat A \to B$ be a functor with small fibers. Consider the induced pair of adjoints:

$$\begin{tikzcd} \mathcal{S}^B \arrow[r, phantom, "\perp"] \arrow[r, bend right, "f^*"'] & \mathcal{S}^A \arrow[l, bend right, "f_!"'] \end{tikzcd}$$

Thus, a square

$$\begin{tikzcd} A' \arrow[r, "u"] \arrow[d, "f'"'] & A \arrow[d, "f"] \\ B' \arrow[r, "v"'] & B \end{tikzcd}$$

induces the *Beck–Chevalley square*

$$\begin{tikzcd} \mathcal{S}^{A'} \arrow[d, "f'_!"'] \arrow[dr, phantom, "\Rightarrow\beta"] & \mathcal{S}^A \arrow[l, "u^*"'] \arrow[d, "f_!"] \\ \mathcal{S}^{B'} & \mathcal{S}^B \arrow[l, "v^*"] \end{tikzcd}$$

with the 2-cell β (a morphism in $\mathcal{S}^A \to \mathcal{S}^{B'}$) being the transpose of the map:

$$u^* \xRightarrow{u^*\eta} u^* f^* f_! \quad \simeq \quad f'^* v^* f_!$$

Analogously to 110, we give a description of the Beck–Chevalley map as in [12, Section 8.4].

Let $F :_\flat A \to \mathcal{S}$ and φ its unstraightening:

$$\begin{tikzcd} X \arrow[r] \arrow[d, "\varphi"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{S}_* \arrow[d] \\ A \arrow[r, "F"] & \mathcal{S} \end{tikzcd}$$

Recall, that $f_! F$ is given by

$$\begin{tikzcd} X \arrow[r, "j"] \arrow[d, "\varphi"'] & Y \arrow[r] \arrow[d, "\psi"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{S}_* \arrow[d] \\ A \arrow[r, "f"] & B \arrow[r, "f_! F"] & \mathcal{S} \end{tikzcd}$$

and that the unit $\eta :_\flat F \to f^* f_! F$ is given by the mediating map:

$$\begin{tikzcd}
X \arrow[r, dashed, "\widetilde{\eta}"] \arrow[rr, bend left, "j"] \arrow[dr, "\varphi"'] & Z \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[d, "\psi"] \\
& A \arrow[r, "f"] & B
\end{tikzcd}$$

Pulling back along u and v, resp., and factoring the upper horizontal map yields the square

$$\begin{tikzcd}
& Y' \arrow[dr, "r"] & \\
A' \times_A X \arrow[ur, "j'"] \arrow[rr, dashed, "j'' := f' \times_f j"] \arrow[d, "u^*\varphi"'] & & B' \times_B Y \arrow[d, "v^*\psi"] \\
A' \arrow[rr, "f'"] & & B'
\end{tikzcd}$$

where j' is initial and r is a covariant fibration. We consider the composite $\psi' := v^*\psi \circ r :_\flat Y' \to B$ whence

$$\begin{tikzcd}
Y' \arrow[r] \arrow[d, "\psi'"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{S}_* \arrow[d] \\
B \arrow[r, "f'_! u^*(F)"] & \mathcal{S}_*
\end{tikzcd}
\qquad
\begin{tikzcd}
B' \times_B Y \arrow[r] \arrow[d, "v^*\psi"'] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[r] \arrow[d, "\psi"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{S}_* \arrow[d] \\
B' \arrow[r, "v"] \arrow[rr, bend right, "v^* f_! F"'] & B \arrow[r, "f_! F"] & \mathcal{S}
\end{tikzcd}$$

and the Beck–Chevalley transformation corresponds to the fibered map $r :_\flat Y' \to_B B' \times_B Y$ over B.

Theorem 111 (Smooth base change, [12, Theorem 8.4.1])**.** *Consider a pullback square*

$$\begin{tikzcd}
A' \arrow[r, "u"] \arrow[d, "f'"'] \arrow[dr, phantom, "\lrcorner", very near start] & A \arrow[d, "f"] \\
B' \arrow[r, "v"'] & B
\end{tikzcd}$$

of small categories where v is smooth. Then, the Beck–Chevalley transformation is an equivalence

$$f'_! u^* \simeq v^* f_!.$$

Доведення. We will show that the map r from the preceding description is an equivalence by showing that it is also initial (and it is a covariant fibration by construction). We have the following commutative cube:

$$\begin{tikzcd}
A' \times_A X \arrow[rr, "j''"] \arrow[dr] \arrow[dd, "u^*\phi"'] \arrow[ddr, phantom, "\lrcorner", very near start] & & B' \times_B Y \arrow[dr, "v'"] \arrow[dd, "v^*\psi"'] \arrow[ddr, phantom, "\lrcorner", very near start] & \\
& X \arrow[rr, "j"] \arrow[dd, "\phi"] & & Y \arrow[dd, "\psi"] \\
A' \arrow[rr, "f'"] \arrow[dr, "u"'] \arrow[dr, phantom, "\lrcorner", very near start] & & B' \arrow[dr, "v"] & \\
& A \arrow[rr, "f"] & & B
\end{tikzcd}$$

By the pullback lemma, the top square is a pullback, too. Since v is smooth and j is initial, j'' is initial. But $j'' = r \circ j'$ so r is also initial due to cancelation. □

One can also prove the analogous *proper* base change formula as in [12, Theorem 8.4.6].

6 Conclusions and future work

We have introduced and studied the impact of the ∞-categorical Yoneda embedding in STT. This includes the development of classical concepts (Kan extensions, adjoints, (co)limits, etc.), all in the synthetic ∞-categorical setting. While some of the basic theory had been investigated in STT already, we were able to produce the first non-trivial concrete examples of, e.g., adjunctions (Theorem 68) and give several more refined versions of existing theorems (Theorem 61) which more closely match their standard counterparts.

6.1 Related work

There are several closely related type-theoretic approaches to synthetic (∞-)category theory. We may roughly divide these into (1) directed type theory, where every type is a category but various operations ($\prod$) must be restricted, and (2) variations on simplicial type theory. For instance, many directed type theories have been proposed and studied over the years [25, 55, 39, 21, 9, 24, 60, 56, 2, 38, 40, 37, ?]. In general, while these type theories are a promising approach to formalize category theory in type theory, none of them have thus far received as much attention as STT and, consequently, none have developed category theory to the extent of this work. Furthermore, it is substantially harder to design a directed type theory in this style (as it is a more radical alteration of the basic rules of type theory) and most proposals handle only 1-category theory rather than $(\infty, 1)$-categories. We note, however, that some of these type theories do include a version of Theorem 62 in the form of directed path induction [38, 40, 37]. Given, however, that few of our arguments rely on types which are not categories, we expect many of them to transfer to sufficiently rich future variants of directed type theory.

Other variations of simplicial type theory have been considered in the literature. For instance, several papers use additional judgmental structure (extension types) to get more definitional equalities around hom-types [46, 4, 57, 10, 59, 58] at the cost of making the interval a second-class type similar to two-level type theory [1, 54]. Other versions have favored a cubical interval [19] or even a cubical interval atop a cubical version of HoTT [60, 56]. Aside from the addition of modalities, our version of STT is deliberately minimalistic: we use only ordinary HoTT with a handful of postulates. Accordingly, our results can be interpret into essentially any incarnation of modal STT and does not rely on extra definitional equalities.

Finally, there are many attempts to formulate more conceptual and synthetic foundations for ∞-category theory which do not rely on type theory. For instance, the ∞-cosmos program of [49] aims to give a systematic account of the formal category theory and model-independence using 2-category theory. On the other hand, most practitioners in the field attempt to give looser "model independent" arguments which avoid relying on explicit computations as much as possible. We have successfully translated some of these arguments into our framework, proving that this informal discipline is effective (e.g., Section 5). More recently, [12] have begun to redevelop ∞-category theory in a deliberately informal and high-level language, splitting the difference between a formal theory like STT and the usual "model-independent" discipline of practitioners. We expect that their arguments can be translated into STT and we have shown that some of their primitive axioms are *provable* in STT (e.g., Theorems 47 and 45 and Lemma 66).

6.2 Future work

Many promising avenues for future work remain to be explored. While we have focused on presheaf categories and immediate consequences of their theory, we plan to port other foundational results from category theory (presentable and accessible categories, Bousfield localizations, topos theory, etc.) into STT. It would also be desirable to adapt more parts of the internal ∞-category theory and ∞-topos theory of Martini and Wolf [29, 35, 31, 34, 36, 30, 61] to STT. Additionally, we hope to extend a proof assistant like Agda [52] with the necessary support for modalities to give machine-checked versions of the proofs in this paper. On the foundational side, STT presently relies on a handful of axioms (Appendix Б) and therefore satisfies only normalization and not canonicity. In future work, we hope to examine which of these principles can be given computational interpretations and to what extent one can 'compute' with synthetic ∞-categories.

A The formal rules of MTT

The formal syntax of MTT is comprised of four judgments: $\vdash \Gamma$, $\Gamma \vdash \delta : \Delta$, $\Gamma \vdash a : A$, and $\Gamma \vdash A$. We list the relevant novel rules for these judgments below:

$$\boxed{\vdash \Gamma}$$

$$\frac{}{\vdash \mathbf{1}} \qquad \frac{\vdash \Gamma}{\vdash \Gamma.\{\mu\}} \qquad \frac{\vdash \Gamma \qquad \Gamma.\{\mu\} \vdash A}{\vdash \Gamma.(\mu \mid A)}$$

$$\frac{\vdash \Gamma}{\vdash \Gamma.\{\mathsf{id}\} = \Gamma \qquad \vdash \Gamma.\{\mu\}.\{\nu\} = \Gamma.\{\mu \circ \nu\}}$$

$$\boxed{\Gamma \vdash \delta : \Delta}$$

$$\frac{}{\Gamma \vdash ! : \mathbf{1}} \qquad \frac{\vdash \Gamma \qquad \Gamma.\{\mu\} \vdash A}{\Gamma.(\mu \mid A) \vdash \uparrow : \Gamma} \qquad \frac{\Gamma \vdash \delta : \Delta \qquad \Gamma.\{\mu\} \vdash a : A}{\Gamma \vdash \delta.a : \Delta.(\mu \mid A)}$$

$$\frac{\Gamma \vdash \delta : \Delta}{\Gamma.\{\mu\} \vdash \delta.\{\mu\} : \Delta.\{\mu\}} \qquad \frac{\vdash \Gamma \qquad \alpha : \mu \longrightarrow \nu}{\Gamma.\{\nu\} \vdash \Gamma.\{\alpha\} : \Gamma.\{\mu\}}$$

$$\frac{\Gamma \vdash \gamma : \mathsf{1}}{\Gamma \vdash\, ! = \gamma : \mathsf{1}} \qquad \frac{\Gamma \vdash \delta : \Delta \qquad \Gamma.\{\mu\} \vdash a : A}{\Gamma \vdash \uparrow \circ (\delta.a) = \delta : \Delta}$$

$$\frac{\Gamma \vdash \delta : \Delta.(\mu \mid A)}{\Gamma \vdash (\uparrow \circ \delta).\mathbf{v}[\delta] = \delta : \Delta.(\mu \mid A)}$$

$$\frac{\Gamma \vdash \delta : \Delta}{\Gamma \vdash \delta.\{\mathsf{id}\} = \delta : \Delta \qquad \Gamma.\{\nu \circ \mu\} \vdash \delta.\{\nu \circ \mu\} = \delta.\{\nu\}.\{\mu\} : \Delta.\{\nu \circ \mu\}}$$

$$\frac{\vdash \Gamma}{\Gamma.\{\mu\} \vdash \Gamma.\{\mathsf{id}\} = \mathsf{id} : \Gamma.\{\mu\} \qquad \Gamma.\{\xi\} \vdash \Gamma.\{\alpha\} \circ \Gamma.\{\beta\} = \Gamma.\{\alpha \circ \beta\} : \Gamma.\{\mu\}}$$

$$\frac{\Gamma \vdash \delta : \Delta \qquad \mu \leq \nu}{\Gamma.\{\nu\} \vdash \Delta.\{\alpha\} \circ \delta.\{\mu\} = \delta.\{\mu\} \circ \Gamma.\{\alpha\} : \Delta.\{\mu\}}$$

$$\frac{\vdash \Gamma \qquad \alpha : \mu_0 \longrightarrow \nu_0 \qquad \beta : \mu_1 \longrightarrow \nu_1}{\Gamma.\{\nu_1 \circ \nu_0\} \vdash \substack{\Gamma.\{\beta\}.\{\mu_0\} \circ \Gamma.\{\alpha\} \\ = \Gamma.\{\beta * \alpha\}} : \Gamma.\{\mu_1 \circ \mu_0\}}$$

$$\boxed{\Gamma \vdash A}$$

$$\frac{\Gamma.\{\mu\} \vdash A}{\Gamma \vdash \mathsf{A}} \qquad \frac{\Gamma \vdash \delta : \Delta \qquad \Delta.\{\mu\} \vdash A}{\Gamma \vdash \mathsf{A}[\delta] = \mathsf{A}[\delta.\{\mu\}]}$$

$$\boxed{\Gamma \vdash a : A}$$

$$\frac{\Gamma.\{\mu\} \vdash A}{\Gamma.(\mu \mid A).\{\mu\} \vdash \mathbf{v} : A[\uparrow.\{\mu\}]} \qquad \frac{\Gamma.\{\mu\} \vdash a : A}{\Gamma \vdash \mathsf{mod}_\mu(a) : \mathsf{A}}$$

$$\frac{\Gamma.(\nu \mid \mathsf{A}) \vdash B \qquad \Gamma.(\nu \circ \mu \mid A) \vdash b : B[\uparrow.\mathsf{mod}_\mu(\mathbf{v})] \qquad \Gamma.\{\nu\} \vdash a : \mathsf{A}}{\Gamma \vdash \mathsf{let}\ \mathsf{mod}_\mu(-) \leftarrow a\ \mathsf{in}\ b : B[\mathsf{id}.a]}$$

$$\frac{\begin{array}{c}\Delta.(\nu \mid \mathsf{A}) \vdash B \\ \Delta.(\nu \circ \mu \mid A) \vdash b : B[\uparrow.\mathsf{mod}_\mu(\mathbf{v})] \qquad \Delta.\{\nu\} \vdash a : \mathsf{A} \qquad \Gamma \vdash \delta : \Delta\end{array}}{\Gamma \vdash \substack{\mathsf{let}\ \mathsf{mod}_\mu(-) \leftarrow a[\delta.\{\nu\}]\ \mathsf{in}\ b[(\delta \circ \uparrow).\mathbf{v}] \\ = (\mathsf{let}\ \mathsf{mod}_\mu(-) \leftarrow a\ \mathsf{in}\ b)[\delta]} : B[\delta.a]}$$

$$\frac{\Gamma.\{\mu\} \vdash a : A \qquad \Gamma \vdash \delta : \Delta}{\Gamma \vdash \mathsf{mod}_\mu(a)[\delta] = \mathsf{mod}_\mu(a[\delta.\{\mu\}]) : \mathsf{A}[\delta.\{\mu\}]}$$

$$\frac{\Gamma \vdash \delta : \Delta \qquad \Gamma.\{\mu\} \vdash a : A[\delta.\{\mu\}] \qquad \Delta.\{\mu\} \vdash A}{\Gamma.\{\mu\} \vdash \mathbf{v}[\delta.a.\{\mu\}] = a : A[\delta.\{\mu\}]}$$

$$\frac{\Gamma.(\nu \mid \mathsf{A}) \vdash B \qquad \Gamma.(\nu \circ \mu \mid A) \vdash b : B[\uparrow.\mathsf{mod}_\mu(\mathbf{v})] \qquad \Gamma.\{\nu\} \vdash a : \mathsf{A}}{\Gamma \vdash (\mathsf{let}\ \mathsf{mod}_\mu(-) \leftarrow \mathsf{mod}_\mu(a)\ \mathsf{in}\ b) = b[\mathsf{id}.a] : B[\mathsf{id}.\mathsf{mod}_\mu(a)]}$$

Б The complete list of axioms

Axiom Б. *Існує множина $\mathbb{I}$, що утворює обмежену дистрибутивну ґратку $(0, 1, \vee, \wedge)$ таку, що $\prod_{i,j:\mathbb{I}} i \leq j \vee j \leq i$ виконується.*

Axiom В. *The map $\mathsf{mod}_\mu(a) = \mathsf{mod}_\mu(b) \to \mathsf{a} = \mathsf{b}$ sending refl to $\mathsf{mod}_\mu(\mathsf{refl})$ is an equivalence for all $a, b :_\mu A$.*

Axiom Г. *There is an equivalence $\neg : \mathbb{I}_{\mathsf{op}} \to \mathbb{I}$ which swaps 0 for 1 and $\vee$ for $\wedge$.*

Axiom Д. *If $A :_\flat \mathcal{U}$, then $\mathsf{A}_\flat \to A$ is an equivalence (A is* discrete*) if and only if $A \to A^{\mathbb{I}}$ is an equivalence (A is* $\mathbb{I}$-null*).*

Axiom Е. *The canonical map $\mathsf{Bool} \to \mathbb{I}$ is injective and induces an equivalence $\mathsf{BoolEquiv}\mathbb{I}_\flat$.*

Axiom Ж. *$f :_\flat A \to B$ is an equivalence if and only if the following holds:*

$$\prod_{n:_\flat \mathsf{Nat}} \mathsf{isEquiv}((f_*)^\dagger : \Delta^{\mathsf{n}} \to \mathsf{A}_\flat \to \Delta^{\mathsf{n}} \to \mathsf{B}_\flat)$$

Axiom И. *For each $n :_\flat \mathsf{Nat}$, there is a (necessarily unique) function $\eta_n :_\flat \Delta^n \to \Delta^{\mathsf{2n+1}}{}_{\mathsf{tw}}$ such that the following map is an equivalence, for each category $C :_\flat \mathcal{U}$:*

$$\iota :\!- \lambda \mathsf{mod}_\flat(f).\, \mathsf{mod}_\flat(f^\dagger \circ \eta_n) : \Delta^{\mathsf{2n+1}} \to \mathsf{C}_\flat \to \Delta^{\mathsf{n}} \to \mathsf{C}_{\mathsf{tw}\flat}$$

Additionally, we require that $\tau = (\mathsf{coe}^\neg)^\dagger : \Delta^{\mathsf{n}}{}_{\mathsf{tw}} \to \Delta^{\mathsf{n}}{}_{\mathsf{op}_{\mathsf{tw}}}$ and that the diagrams in Figure 3 commute (these are mere properties—all objects are sets since $-_\mu$ preserves h-level).

The following *duality axiom* was first studied by [8] and implies that, e.g., $\mathbb{I}$ is a category. Closely related axioms and consequences are considered by [41] and [11]. We did not introduce it in the main body of the paper as it was not explicitly invoked in any of our proofs.

Axiom K. *If A is a finitely presented $\mathbb{I}$-algebra (i.e., A is a bounded distributive lattice equivalent to $\mathbb{I}[x_1, \ldots, x_n]$ quotiented by finitely many relations) and $\mathsf{hom}_{\mathbb{I}\mathrm{Alg}}(A, \mathbb{I})$ is the type of $\mathbb{I}$-algebra homomorphisms, then the map $\lambda a\, f.\, f(a) : A \to (\mathsf{hom}_{\mathbb{I}\mathrm{Alg}}(A, \mathbb{I}) \to \mathbb{I})$ is an equivalence.*

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